



Review

Microstructural and compositional design of Ni-based single crystalline superalloys — A review

Haibo Long ^{a, **}, Shengcheng Mao ^{a, **}, Yinong Liu ^{b, ***}, Ze Zhang ^{a, c}, Xiaodong Han ^{a, *}^a Institute of Microstructure and Property of Advanced Materials, Beijing Key Lab of Microstructure and Property of Advanced Materials, Beijing University of Technology, 100124 Beijing, China^b School of Mechanical and Chemical Engineering, The University of Western Australia, Perth, WA 6009, Australia^c State Key Laboratory of Silicon Materials and Department of Materials Science and Engineering, Zhejiang University, 310058 Hangzhou, China

ARTICLE INFO

Article history:

Received 6 November 2017

Received in revised form

15 January 2018

Accepted 16 January 2018

Available online 31 January 2018

Keywords:

Superalloy

Lattice misfit

TCP phase

Precipitation

ABSTRACT

The microstructure, thermal and mechanical stability of Ni-based single crystalline superalloys depend strongly on the alloying elements and their concentrations. Alloying has been the main design strategy for stabilizing the compositions, microstructures and thermal-mechanical properties. This article presents a review on the effects of some common alloying elements on the microstructural and mechanical property stability control of Ni-based superalloys. The various alloying elements are divided into four categories according to their main effects on these properties, comprising base elements, mechanical strengthening elements, long term stability elements and the oxidation resistance elements. The mechanical strengthening elements can further be divided into precipitation, solid solution and grain boundary segregation elements. The precipitation elements strengthen the alloys by forming the L1₂ structured γ' phase. The solid solution elements strengthen primarily the γ phase, by increasing the solidus temperatures and decreasing the stacking fault energy, which in turn influences the thermal stability of the phases and the resistance of dislocation movement. The grain boundary elements strengthen the alloys by the formation of carbides and borides along the grain boundaries during solidification, which help to prevent the formation of casting pores and hot tearing and to strengthen low angle boundaries. The long-term stability elements inhibit the precipitation of topologically closed-packed phases causing deterioration of the mechanical properties. The oxidation resistance element, mainly Al, promotes the formation of protective Al₂O₃ surface layer.

© 2018 Elsevier B.V. All rights reserved.

Contents

1. Introduction	204
2. Chemical compositions of Ni-based single crystalline superalloys	204
2.1. Evolution of Ni-based single crystalline superalloys	204
2.2. Classification of alloying elements in Ni-based single crystalline superalloys	206
3. Mechanical strengthening elements	207
3.1. Microstructure of Ni-based superalloys	207
3.2. Precipitation elements	207
3.2.1. Optimal morphology of the γ' phase	207
3.2.2. Strength mechanism of the γ' phase	208
3.2.3. Effect of alloying elements on strengthen of the γ' phase	209
3.3. Solid solution elements	209
3.3.1. Solid strengthen effect	210

* Corresponding author.

** Corresponding author.

*** Corresponding author.

E-mail address: xdhan@bjut.edu.cn (X. Han).

3.3.2.	Stacking fault energy	210
3.3.3.	Melting temperature	211
3.3.4.	Other effect	212
3.3.5.	TCP phase formation	212
3.4.	Grain boundary elements	213
3.5.	Lattice misfit	213
3.5.1.	Effects of alloying elements on lattice misfit	213
3.5.2.	Influence of lattice misfit on γ' morphology	213
3.5.3.	Influence of lattice misfit on dislocation network at the γ'/γ interface	215
3.6.	Long term stability elements	215
4.	Oxidation resistance elements	216
5.	Summary	216
	Acknowledgement	217
	References	217

1. Introduction

Superalloys are the materials of choice for components serving under load at elevated temperatures above 540 °C, because of their excellent ability to retain strength, to withstand creep and fatigue, and to resist oxidation at such temperatures [1–3]. One of the most common applications of superalloys is turbine blades, which operate at high temperatures and under constant centrifugal forces [4,5]. There are three types of superalloys, i.e., the Fe-based, Ni-based and Co-based alloys. Among these three types superalloys, Ni-based alloys are the most widely used due to their more superior high temperature properties relative to the Fe-based superalloys and their better affordability relative to the Co-based superalloys [6–9].

Ni-based superalloys derive their high temperature strength from a high volume fraction γ' phase, which has an ordered $L1_2$ structure. The γ' phase exists in the alloy in the form of cuboids with coherent interfaces with the continuous fcc γ phase matrix, which presents as narrow channels between the γ' cuboids. The highly ordered $L1_2$ structure renders the γ' phase high rigidity and low dislocation tolerance, thus the dislocation movement are mainly limited in the γ channels. This provides the high-temperature strength and creep resistance of these alloys. When serving at harsh thermal-mechanical environments, three major microstructure evolution will happen, i.e., formation of dislocation networks, coarsening of the γ' phase cuboids, and precipitation of topologically closed packed (TCP) phase. The last two types of microstructural evolution are harmful to the creep resistance and they reflect the intrinsic microstructural instability of the alloys. The formation of dislocation networks has dual effects. On one hand, dislocation networks can improve creep resistance in a mechanism similar to strain hardening. On the other, high density dislocations in a network promote diffusion and coarsening of the γ' phase cuboids. Generally, two strategies have been taken to improve creep resistance of these alloys, i.e., microstructure control and composition design.

Superalloys have gone through a continuous development since the design of the first Ni-based superalloy in the 1940s [10–12]. To date, we have seen four major stages of microstructural development of superalloys [13–15]. The first superalloys were produced in wrought forms, like most other metal products. These products, particularly for turbine blades, which have complex surface profiles and are produced primarily by computer numerical control machining, suffered from the problems of high production cost, low success rate and high material wastage [16,17]. The second stage saw the development of cast superalloys in the 1950s, which offered great advantages in high production throughput and low

cost, low material wastage and improved high temperature strength and creep resistance [18,19]. Further development was achieved with the production of columnar grained superalloys via directional solidification casting [20–22]. This technique is clearly targeted for turbine blade applications where the uniaxial mode of loading in service is defined. Grain boundaries are weaker parts of the alloy for high temperature strength applications, by acting as nucleation sites for micro cracking and by experiencing grain boundary sliding in shear motion. The columnar grained microstructure features grain boundaries only in the direction parallel to the tensile centrifugal load and eliminates grain boundaries in all other orientations that may experience a shear component from the load, thus significantly improved the high temperature strength and creep resistance of the alloys. The last major improvement in microstructural design is the production of single crystalline superalloys, produced by solidification control. By eliminating grain boundaries all together, single crystalline superalloys achieved the highest high temperature mechanical strength and creep resistance possible by microstructural design [23–25]. Fig. 1 shows an overview of the improvement of creep service life at elevated temperature of Ni-based superalloys with the continued development of microstructures.

With the exhaustion of potential for property improvement by means of microstructure design, further improvement requires a different approach. The second approach to improving the creep resistance and service life of superalloys is composition design. Alloying elements in superalloys may affect many characteristics, e.g., stacking fault energy, antiphase boundary energy, critical resolved shear stress for dislocation movement, grain boundary stability, volume fractions of γ' precipitates, and lattice misfit. A large number of alloying elements, including Co, Cr, Mo, W, Al, Ti, Ta, Re and Ru, have been used in Ni-based superalloys for various purposes, such as to improve oxidation resistance, to increase mechanical strength and to enhance long-term microstructural stability [26–38].

This review summarizes the various effects of alloying elements on the microstructure, mechanical properties and chemical characteristics of Ni-based single crystalline superalloys, with a purpose for better understanding for element selection in superalloy design.

2. Chemical compositions of Ni-based single crystalline superalloys

2.1. Evolution of Ni-based single crystalline superalloys

The first single crystalline Ni-based superalloy was PWA1480 produced in the 1980s [39,40]. Since then several series of Ni-based

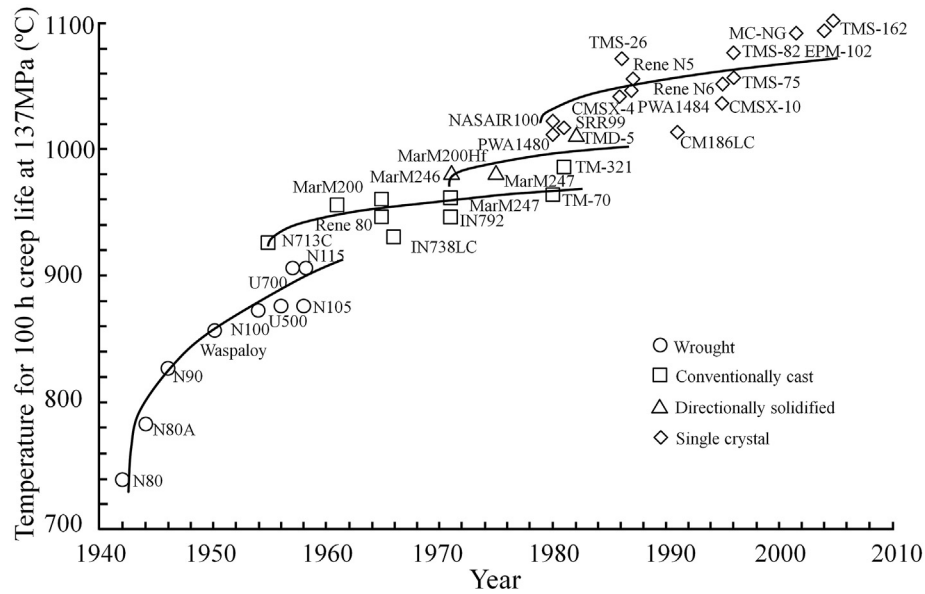


Fig. 1. History of endurance temperature improvement with respect to microstructural development of Ni-based superalloys [10].

single crystalline superalloys of different chemical compositions have been designed to improve the mechanical and chemical properties. One criterion for the evaluation of high temperature performance of superalloys is the highest temperature at which the alloy can endure creep testing for 1000 h at 137 MPa [10]. Based on this criterion, the various Ni-based single crystalline superalloys developed to date are generally classified into six generations.

Fig. 2 shows the endure temperature of some typical Ni-based single crystalline superalloys [10,41–43]. The endure temperature of superalloys increased from ~1000 °C for the 1st generation alloys to ~1120 °C for the 6th generation superalloys. Table 1 [44–48] lists the compositions of the Ni-based single crystalline superalloys shown in Fig. 1. The Ni-based single crystalline superalloys mainly

contain Ni, Co, Cr, Mo, W, Al, Ti, Ta, Re and Ru and a small amount of some other elements such as C, B, and Hf.

From the viewpoint of chemical composition, the main difference between the 1st and the 2nd generations is the addition of Re. The 3rd generation was developed by increasing the Re content from 3.0 wt.% to >5.0 wt.%. The 4th generation saw the addition of Ru and the 5th generation further increased the Ru content from 3.0 wt.% to >5.0 wt.% and adjusted for the optimal Re/Ru ratio. The 6th generation was created by fine tuning the contents of other elements.

Re is an effective strengthening element, with ~30 °C increase in endure temperature for 3 wt.% Re addition, as observed in the first three generations of superalloys [49,50]. However, excessive

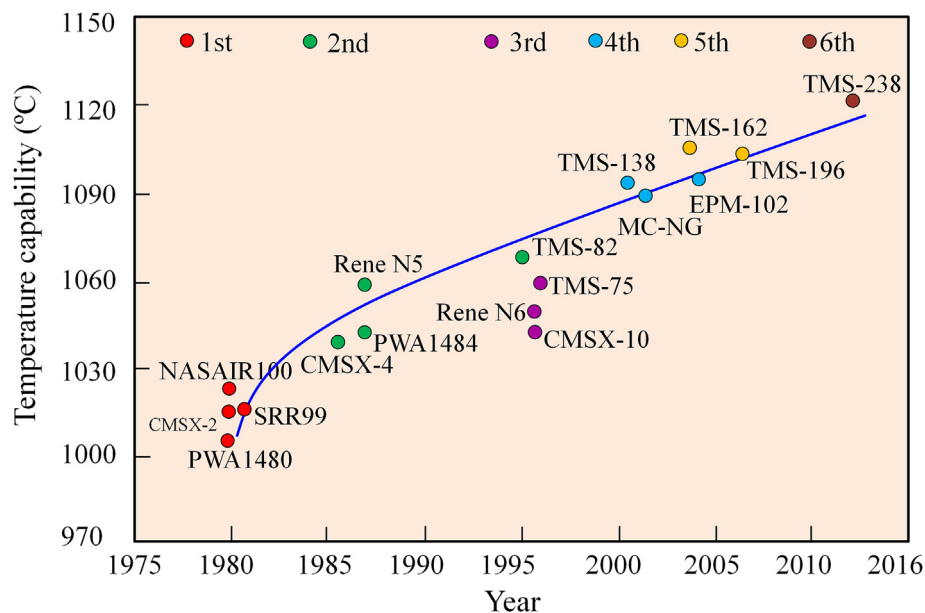


Fig. 2. High temperature capability of some typical Ni-based single crystalline superalloys [10,41–43]. The high temperature capability is evaluated by the highest temperature at which an alloy can endure for 1000 h of creep testing under 137 MPa tensile stress without fracture.

Table 1

Compositions of some typical Ni-based single crystalline superalloys (wt.%) [44–48].

Generation/Alloy	Ni	Co	Cr	Mo	W	Al	Ti	Ta	Re	Ru	Else
1st	PWA1480	Bal.	5.0	10.0	—	4.0	5.0	1.5	12.0	—	—
	CMSX-2	Bal.	4.6	8.0	0.6	8.0	5.6	1.0	9.0	—	—
	SRR99	Bal.	5.0	8.0	—	10.0	5.5	2.2	12.0	—	0.015C
	NASAIR100	Bal.	—	8.5	1.1	10.1	5.8	1.2	3.3	—	—
2nd	PWA1484	Bal.	10.0	5.0	2.0	6.0	5.6	—	9.0	3.0	0.1Hf
	Rene N5	Bal.	8.0	7.0	2.0	5.0	6.2	—	7.0	3.0	0.2Hf
	CMSX-4	Bal.	9.0	6.5	0.6	6.0	5.6	1.0	6.5	3.0	0.1Hf
3st	Rene N6	Bal.	12.5	4.2	4.4	6.0	5.7	—	7.2	5.4	0.05C
											0.004B
											0.15Hf
	CMSX-10	Bal.	3.0	2.0	0.4	5.0	5.7	0.2	8.0	6.0	0.03Hf
	TMS-75	Bal.	12.0	3.0	2.0	6.0	6.0	—	6.0	5.0	0.1Hf
4th	PWA1497	Bal.	16.5	2.0	2.0	6.0	5.5	—	8.2	5.9	0.15Hf
											0.03C
	MC-NG	Bal.	—	4.0	1.0	5.0	6.0	0.5	5.0	—	0.1Hf
	EPM-102	Bal.	16.5	2.0	2.0	6.0	5.5	—	8.3	5.9	0.15Hf
											0.03C
5th	TMS-162	Bal.	5.8	3.0	3.9	5.8	5.8	—	5.6	4.9	0.09Hf
	TMS-196	Bal.	5.6	4.5	2.4	5.0	5.6	—	5.6	6.4	0.1Hf
6th	TMS-238	Bal.	6.5	4.6	1.1	4.0	5.9	—	7.6	6.4	0.1Hf

amount of Re breaks the solid solution equilibrium in local regions, leading to the precipitation of TCP phases, which can significantly decrease the creep property of superalloys because of the following reasons. First, precipitation of the TCP phase consumes the strengthening elements in the matrix, thus decreases the strengthening effect. Second, the TCP phases are intrinsically brittle, thus are easy to fracture under external load. Thirdly, the excessively large lattice misfits between the TCP phases and the γ' phase produce high internal stresses at the interface and make it a preferred region for crack nucleation, thus seriously deteriorates the creep life in the 3rd generation superalloys [51,52]. To inhibit the precipitation of TCP phases, 3 wt.% Ru was added in the 4th generation alloys, achieving an endure temperature enhancement from $\sim 1060^\circ\text{C}$ for the 3rd generation superalloys to $\sim 1080^\circ\text{C}$ for the 4th generation superalloys [27,28]. The 5th and 6th generation superalloys adjusted the proportion of the Re, Ru and some other elements for better collocation.

2.2. Classification of alloying elements in Ni-based single crystalline superalloys

The alloying elements in Ni-based single crystalline superalloys can be categorized into four types based on their main contributions to the alloys, i.e., (i) the base elements (i.e., Ni for Ni-based superalloys), (ii) the mechanically strengthening elements, (iii) the long term stability elements and (iv) the oxidation resistance elements. This classification is schematically illustrated in Fig. 3. Some of the elements may have multiple effects and thus belong to more than one category. For example, Al and Cr may both strengthen the alloy matrix and improve oxidation resistance, and Ru both mechanically strengthens the alloy and stabilizes its microstructure against precipitation of TCP phases. The mechanical strengthening elements can be further divided into three sub-groups, including the precipitation strengthening elements (Al, Ta, and Ti), grain boundary strengthening elements (C and B) and solid

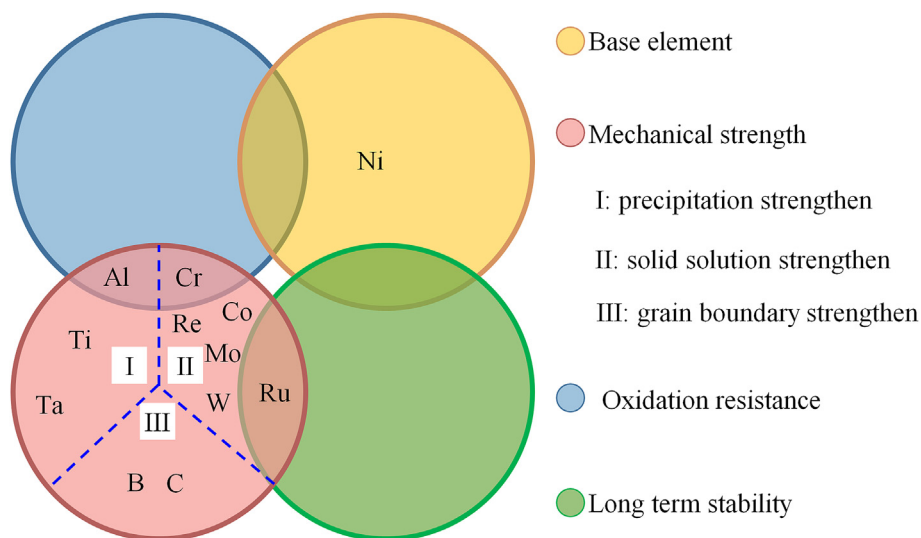


Fig. 3. Classification of the alloying elements in Ni-based superalloys. The yellow, blue, red and green circle represent the base element, the oxidation resistance element, the mechanical strength element and long term stability elements respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

solution strengthening elements (Cr, Co, Mo, W, Re, and Ru). The effects of these elements on the microstructure and properties of superalloys are discussed below.

3. Mechanical strengthening elements

3.1. Microstructure of Ni-based superalloys

The microstructure of Ni-based single crystalline superalloys is usually a dual phase microstructure consisting of an ordered L1₂ structured γ' phase and a FCC structured γ phase, as shown in Fig. 4. Fig. 4(a) and (b) are typical micrographs of the as-cast and the precipitation-aged γ'/γ composite microstructures of a Ni-based single crystalline superalloy respectively [53]. The dual phase composite structure is the “service” microstructure of the alloys and is formed by γ' precipitation via post-casting heat treatment. The γ' precipitates exist in cuboidal shapes with {100} facets and all aligned in $\langle 100 \rangle$ directions of the single crystalline, leaving the γ matrix in narrow parallel channels in between the γ' cuboids. The two phases share perfectly coherent interfaces [9,14,54].

The L1₂ γ' phase and the FCC γ phase have different chemical compositions. Being an ordered structure, the L1₂ γ' phase has a higher priority, or selectivity, of elements. The stoichiometry of the L1₂ structure is A₃B, and a typical example is Ni₃Al, as schematically shown in Fig. 4(c). In one γ' phase unit cell three Ni atoms as the host element are expected to occupy the face-centered positions and one atom of the alloying elements is expected to occupy the eight apex angle positions [55,56]. In the event that not a single alloying element is sufficient to fulfil the apex angle positions of the cubic unit cell, other elements join the occupancy to form “alloyed” γ' phases. The same also occurs for the Ni atoms at the face-centered sites. Being a solid solution structure, the γ phase has low preference of the alloying elements. It has been reported that Ni, Al, Ti and Ta prefer to segregate into the γ' phase, whereas Cr, Co and Re prefer to congregate into the γ phase [57–60]. To evaluate the phase preference of elements, γ' phase segregation coefficient is defined as the ratio of the concentration of an element in γ' to that in γ , i.e., $k_{\gamma'} = \frac{C_{\gamma'}}{C_{\gamma}}$.

3.2. Precipitation elements

Precipitation elements refer to the elements that have a higher concentration in the γ' phase than in the γ phase, i.e., having a segregation coefficient of $k_{\gamma'} = \frac{X_{\gamma'}}{X_{\gamma}} > 1$. The γ' phase is the dominant

strengthening strategy of Ni-based superalloys. It strengthens the alloy primarily by two mechanisms, i.e., precipitation strengthening by hindering the propagation of dislocations in the γ phase matrix, and structural strengthening by having an ordered crystal structure which is less tolerant to dislocations [61,62]. Both are effective in lowering creep rate and extending service life [63,64]. The common precipitation elements in Ni-based superalloys, in addition to Ni and Al, are Ti and Ta [30,65–68].

3.2.1. Optimal morphology of the γ' phase

The creep performances of Ni-based single crystalline superalloys have been found to strongly correlate to the morphology of the γ' phase, as shown in Fig. 5. Fig. 5(a) shows a comparison of the minimum creep rates among alloys with different phase structures, including γ , γ' and $\gamma + \gamma'$ [69], the compositions of alloys 1–7 are listed in Table 2 [70–73]. The minimum creep rates of all these three types of superalloys increase proportionally with increasing the applied stress. Under the condition of the same applied stress, however, the minimum creep rate appears to increase in the order of $\gamma + \gamma' < \gamma' < \gamma$. This demonstrates that precipitation of the γ' phase in the γ matrix can significantly improve the creep resistance. Fig. 5(b) shows the effect of the γ' phase volume fraction on creep rupture life of TMS-75 and TMS-82 [10,74]. The creep life shows a peak value at the volume fraction range of 60%–75% for both alloys. Fig. 5(c) shows the relationships between the creep rupture life and the size and shape of the initial primary γ' phase [75]. Different γ' precipitate morphologies were evaluated, including spherical, cuboidal and irregular raft structures. The cuboidal morphology exhibits the highest creep life among all morphologies and the optimal initial size of the cuboids is $\sim 0.5 \mu\text{m}$.

To achieve high creep resistance performance, the commercial Ni-based single crystalline superalloys generally are designed to have a microstructure consisting of 60%–75% volume fraction of the γ' phase, cuboidal morphology and initial size of 0.2–0.6 μm in cube edge length at the room temperature. The morphology and initial size can be achieved by heat treating the alloys through a solution treatment and a two-stage aging process. The two-stage aging treatment is designed to achieve a desired cubicity and size uniformity of the γ' phase [76–79]. The volume fraction of the γ' phase can be controlled by adjusting the amount of precipitation elements.

The Ni–Al binary diagram has been traditionally used to tune the Al content to optimize the volume fraction of the γ' phase. The solubility of Al in Ni is about 15 at.% at 1100 °C [10,80–82]. The stoichiometric content of Al in Ni₃Al is 25 at.%. To obtain a desired γ'

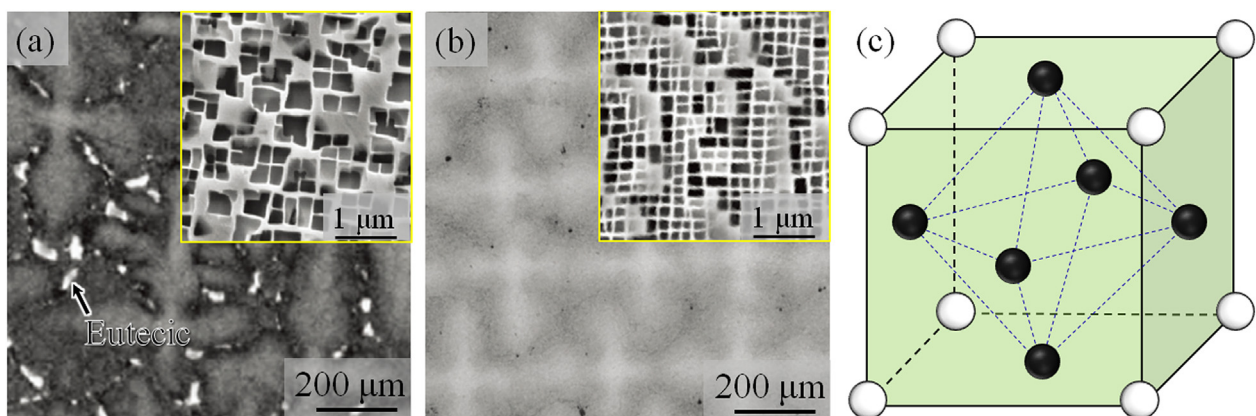


Fig. 4. Microstructures of Ni-based single crystalline superalloys [54]. (a) Microstructure of single crystalline DD22 superalloy in as-cast state; (b) Microstructure of the DD22 superalloy after γ' phase formation heat treatment; (c) Schematic of the A₃B type L1₂ structure of the γ' phase. The insets in (a) and (b) are the γ - γ' microstructure at higher magnifications.

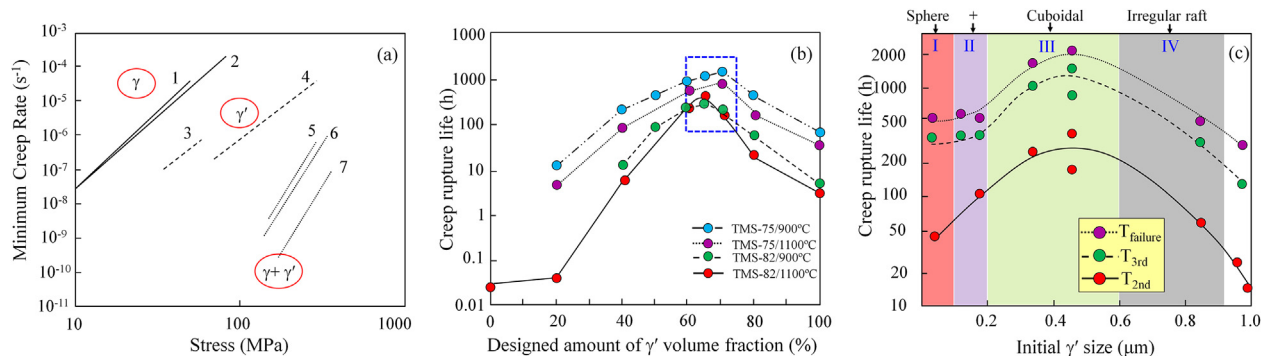


Fig. 5. Effects of γ' phase morphology on the creep performance of Ni-based single crystalline superalloys. (a) Comparison of the minimum creep rates of Ni-based single crystalline superalloys of different phase structures of γ , γ' and $\gamma + \gamma'$ [69]. Compositions of alloys 1–7 are listed in Table 2. (b) Effect of γ' phase volume fraction on creep rupture life of two Ni-based single crystalline superalloys [74]. (c) Effect of the initial size of γ' cuboids on creep rupture life of MAR-M200 superalloy. The three curves represent the onset times for the second stage creep, third stage creep and the time of final rupture [75].

Table 2

Alloy identification of the seven superalloys presented in Fig. 5(a) [70–73].

	Alloy composition	microstructure
1	Ni–2W (wt.%)	γ
2	Ni–26.5Cr–4.4Al–3.3W–0.6Mo–18.3Co (at.%)	γ
3	Ni–12.4Al (wt.%)	γ'
4	Ni–2.2Cr–18.8Al–2.4Ti–1.7Ta–1.5W–5.7Co (at.%)	γ'
5	NASAIR 100	$\gamma + \gamma'$
6	Rene N4	$\gamma + \gamma'$
7	Alloy SX-1	$\gamma + \gamma'$

volume fraction (V in %), the content of Al (x in at.%) in the alloy needs to satisfy $(x - 15)/(25 - 15) = V$. Thus for a desired volume fraction of 60–75% of the γ' phase, the Al content is estimated to be 21–22.5 at.% (or 11–12 wt.% Al). As can be seen from Table 1, the Al contents in commercial Ni-based single crystalline superalloys are ~6 wt.%, well below what may be required to form a pure Ni_3Al γ' phase structure of 60–75% in volume fraction. The shortfall of Al is compensated by the addition of Ti and Ta.

3.2.2. Strength mechanism of the γ' phase

As stated above, the γ' phase is less tolerant to dislocations. As a result, it has high strength and also it hinders the movement and propagation of dislocations inside the γ phase. The strength of the γ' phase, or its low tolerance for dislocations, derives from its L_{12} ordered structure. In an L_{12} structure, the Burger's vector of a full dislocation is $b = a < 110 >$, which is twice in magnitude of that in the FCC structure. Fig. 6 shows the $[111]$ projection of three neighboring (111) planes in the L_{12} structure. The red lattice points represent the Ni atoms and the blue represent the Al atoms in the Ni_3Al structure. The three different sizes of the lattice points represent the three consecutive (111) atomic layers. It is apparent that a shear displacement of the top (111) plane by a vector of $b = a/2[\bar{1}01]$, which is the Burger's vector of a full dislocation in the FCC γ phase, an anti-phase boundary (APB) is created between the two adjacent (111) planes. This converts Ni–Al bonds in the L_{12} structure to Ni–Ni and Al–Al bonds. To fully restore the perfect L_{12} ordering, the shear displacement must continue to $b = 2 \times b = a[\bar{1}01]$, which defines the full dislocation in the L_{12} structure. This explicitly demonstrates that a full dislocation in the γ' phase is equivalent to two full dislocations in the γ phase and that it has a much higher lattice distortion energy (by a factor of 4 because the lattice distortion energy is a quadratic function of the magnitude of the Burger's vector). This makes the creation,

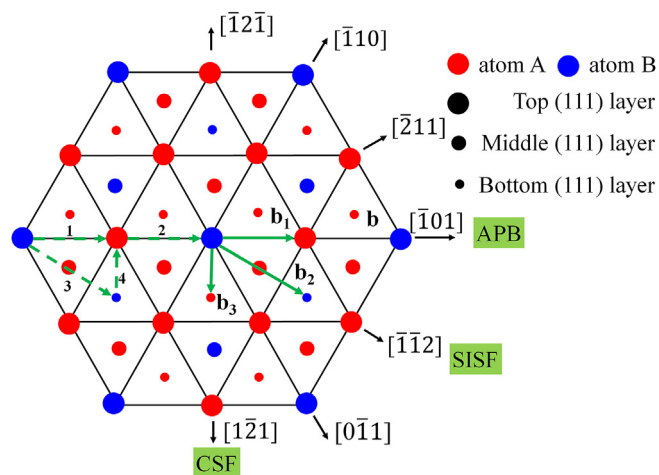


Fig. 6. Schematic of $[111]$ direction projection of the Ni_3Al structure and dislocation formation and movement mechanisms in the structure, and the planar defects created by dislocation movements. The projection contains three successive $\{111\}$ planes of Ni_3Al , represented by the different sizes of the lattice points. The red points represent the Ni sites (atom A) and the blue points represent the Al sites (atom B). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

presence and movement of dislocations in the γ' phase much more difficult.

To overcome this prohibitively high energy barrier, dislocations from the γ matrix always propagate into the γ' phase in pairs, with each functions as a partial dislocation in the γ' phase. There are two basic mechanisms by which dislocation pairs from the γ phase can enter the γ' phase [83,84]. In mechanism I, a $b = a/2[\bar{1}01]$ dislocation and a dislocation slip consecutively on the same (111) plane into the γ' phase. The leading dislocation creates an APB on its path, which leads to an increase in system energy of ~150 mJ/m² in Ni–Al alloy [85], and the trailing dislocation eliminates the APB and restores the L_{12} ordering of the matrix. This is expressed as $1 + 2$ in Fig. 6.

In mechanisms II, a $b = a/3[\bar{1}12]$ dislocation and a $b = a/6[\bar{1}21]$ dislocation may slip as a pair into the γ' phase across the same (111) plane. The passing of a $b = a/3[\bar{1}12]$ dislocation changes the stacking sequence of (111) planes from ABCABCABC to ABCACABC locally, forming a super lattice intrinsic stacking fault (SISF). The nearest Ni–Al bond is not affected by this shear process, resulting

in low SISF energy of $\sim 10 \text{ mJ/m}^2$ in Ni–Al alloy [10]. On the other hand, the passing of a $b = a/6[1\bar{2}1]$ dislocation by itself creates a complex stacking fault (CSF). The CSF has a high formation energy of $\sim 250 \text{ mJ/m}^2$ in the Ni–Al alloy, because more Ni–Al bonds are converted to Ni–Ni and Al–Al bonds [86]. However, if a $b = a/6[1\bar{2}1]$ dislocation follows a $b = a/3[1\bar{1}2]$ dislocation into the γ' phase, it turns the SISF into an APB instead of forming its own CSF, and an APB is more energy saving than a CSF. Therefore, $b = a/6\langle 112 \rangle$ dislocations prefer to follow the SISF forming $b = a/3\langle 112 \rangle$ dislocations. This is expressed as $3 + 4$ in Fig. 6.

In summary, it is apparent above that dislocation movement in the γ' phase requires to overcome a minimum energy equaling to the formation energy of APB. This is a net system energy increase in addition to the dissipation energy associated with the breaking and reforming of atomic bonds during the movement of dislocation movement, which is also experienced by the γ phase. Furthermore, it is also clear that plasticity of the γ' phase requires the coordinated movement of two dislocations. All these provide effective strengthening mechanisms for the γ' phase.

3.2.3. Effect of alloying elements on strengthen of the γ' phase

The above also allows us to think that alloying elements in the γ' phase, which inevitably affect the formation energies of APB, SISF and CSF, may also directly influence the strength and plasticity of the γ' phase. Indeed, in commercial Ni-based single crystalline superalloys, the strengthening effect of the γ' phase is further improved by addition of Ta and Ti. This is because that Ta–Ta and Ti–Ti have higher bonding energies than does Al–Al bond.

Fig. 7(a) shows the dependence of APB energy on Ti/Al ratio in Ni–Al system [85]. It is seen that substitution of Al by Ti increases the APB energy, from $\sim 150 \text{ mJ/m}^2$ in the pure Ni–Al system (Ti/Al = 0) to ~ 240 (Ti/Al = 1) and $\sim 300 \text{ mJ/m}^2$ (Ti/Al = 8) in Ni–Al–Ti alloys, with a maximum net increase of APB energy of $\sim 150 \text{ mJ/m}^2$ (Ti/Al = 8). An increase in the APB energy implies an increase in the resistance to dislocation movement in the ordered structure, thus further strengthening of the γ' phase and further prolonged rupture life [87,88]. However, excessive addition of Ti also has two main negative impacts. First, it increases the positive misfit between the γ' and γ phases, thus increasing the coarsening rate of the γ' phase in service [88–91]. Second, a high Ti/Al ratio promotes the

precipitation of a η -Ni₃Ti phase. The η -Ni₃Ti phase has a tetragonal structure with $a = b = 5.09 \text{ \AA}$, and $c = 8.29 \text{ \AA}$ [92,93]. The η -Ni₃Ti/ γ interface is a vulnerable site for crack nucleation because of the significant difference in lattice structure between the two phases. These two negative effects can decrease the creep rupture life of the superalloys. It is also evident in Fig. 7(a) that the increase of APB energy is most rapid at Ti/Al ratios up to 1. The potency of Ti addition within this range may be estimated to be 90 mJ/m^2 per Ti/Al = 1. The optimum Ti/Al ratio for maximum rupture life is determined to be 1.22 [59].

Fig. 7(b) shows the effect of Ta/Al ratio on APB energy [65]. It is seen that an increase in the Ta/Al ratio may cause an immediate and significant increase of the APB energy to $\sim 600 \text{ mJ/m}^2$ at a low Ta/Al ratio of ~ 0.2 , followed by a continuous decrease to below 0 when the ratio is close to 5. A nil APB energy implies that there is no preference of an ordered L1₂ structure to a disordered FCC structure. Based on this knowledge, Ta is always added in small amount to increase the APB energy. The potency of Ta addition on APB energy reduction, at Ta/Al < 0.2, may be estimated to be 1500 mJ/m^2 per Ta/Al = 1, and the next increase of APB energy caused by Ta addition is $\sim 400 \text{ mJ/m}^2$. In comparison, Ta is a much more effective and efficient strengthening element than Ti [94–97].

Table 3 lists the Ti/Al and Ta/Al molar ratios of some commercial Ni-based single crystalline superalloys. The Ti/Al ratios in all the commercial superalloys are lower than 1 and tend to decrease in trend with the newer generations. This tendency may be attributed to the relative ineffectiveness of Ti in increasing the APB energy, and more importantly the two negative effects explained above. The Ta/Al ratios have been tuned to be within the range of 0.15–0.35, in accordance with the findings presented in Fig. 7(b).

3.3. Solid solution elements

These elements refer to those whose primary function in Ni-based superalloys is solid solution strengthening of the γ phase. These elements include Cr, Co, Mo, W, Re and Ru. In addition to causing solid solution strengthening, these elements may also have other effects, such as increasing the solidus temperatures and decreasing the stacking fault energy (SFE), which in turn influences the thermal stability of the phases and the resistance of dislocation

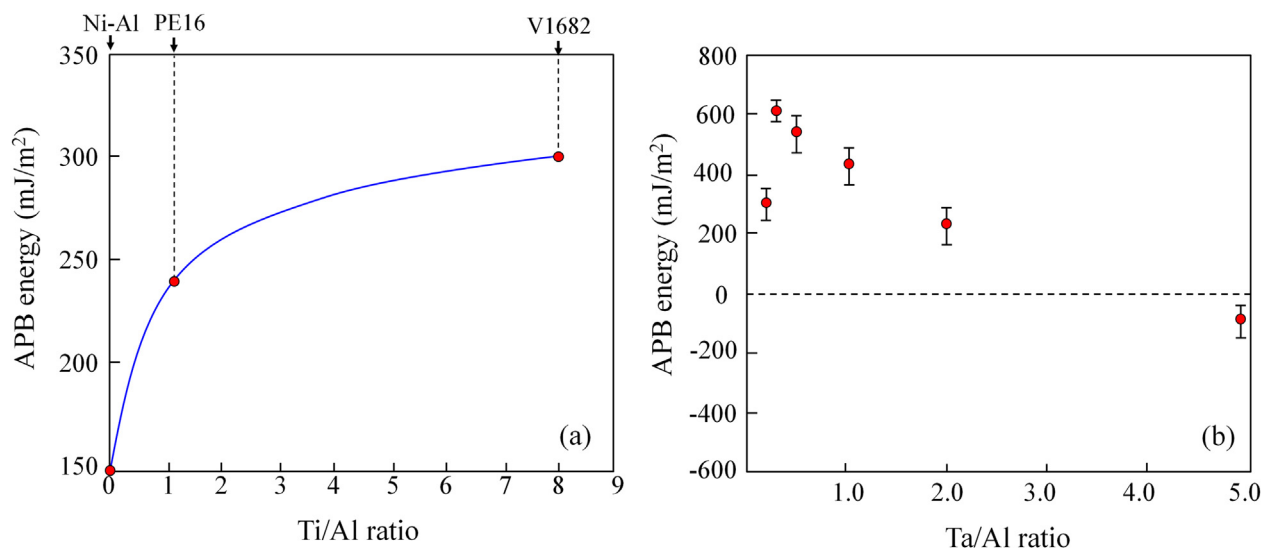


Fig. 7. Effects of Ti and Ta on anti-phase boundary (APB) energy of Ni-based alloys. (a) Dependence of APB energy on Ti/Al ratio of three Ni-based alloys [85]. (b) Dependence of APB energy on Ta/Al ratio of Ni₃Al_(1-x)Ta_x as determined by using molecular dynamic simulation [65].

Table 3
Ti/Al and Ta/Al ratios in the commercial Ni-based single crystalline superalloys listed in Table 1 (at.%) [44–48].

Generation/Alloy	Al	Ti	Ta	Ti/Al	Ta/Al
1st	PWA1480	11.26	1.90	4.03	0.17
	CMSX-2	12.69	1.28	3.04	0.10
	SRR99	12.86	2.90	4.18	0.23
2nd	NASAIR100	12.80	1.49	1.09	0.12
	PWA1484	12.95	0.00	3.10	0.00
	Rene N5	13.90	0.00	2.34	0.00
3st	CMSX-4	12.59	1.27	2.18	0.10
	Rene N6	13.34	0.00	2.51	0.00
	CMSX-10	13.21	0.26	2.76	0.02
4th	TMS-75	13.75	0.00	2.05	0.00
	PWA1497	13.22	0.00	2.94	0.00
	MC-NG	13.00	0.61	1.61	0.05
5th	EPM-102	13.23	0.00	2.98	0.00
	TMS-162	13.85	0.00	1.99	0.00
	TMS-196	13.29	0.00	1.98	0.00
6th	TMS-238	13.97	0.00	2.68	0.00

movement.

3.3.1. Solid strengthen effect

The solid solution elements strengthen the γ matrix mainly by increasing the resistance to dislocation movement. This is because that all the solute atoms identified above have larger atomic diameters than the Ni, thus can distort the lattice and increase the shear moduli [47,98–102]. Their effectiveness is limited by their solubilities in the γ matrix. The Hume-Rothery rule states that a high solubility is achieved when the deviation of atomic diameter is less than $\pm 15\%$ [103–105].

Fig. 8(a) presents the deviation of atomic diameters of all the alloying elements from that of Ni in Ni-based superalloys [10,106]. It is seen that the atomic diameters of all these elements are larger than that of Ni and that the deviations all fall within the range of 0–15%. The solubility limits in at.% of these elements are also indicated in the figure.

Fig. 8(b) shows the dependence of the yield strength on temperature of three Ni-based superalloys (U720LI alloys) containing different amounts of Co [107]. It is seen that higher Co content increases the yield strength. In addition to yield strength, solid

solution elements can also improve the ultimate tensile strength [72,99,108].

3.3.2. Stacking fault energy

The stacking fault energy (SFE) is the additional energy required for the formation of a stacking fault of unit area. Fig. 9 illustrates schematically the effect of SFE in FCC alloy. The full dislocation $b = a/2 < 110 >$ in FCC may decompose into two partial dislocations following reaction $a/2[110] \rightarrow a/6[121] + a/6[21\bar{1}]$, as in the $(1\bar{1}1)$ gliding plane as an example. Fig. 9(a) illustrates the situation when a screw dislocation dissociates into two 30° mixed dislocations via such reaction. A stacking fault is created by the passing of the leading $b_2 = a/6[121]$ partial dislocation and eliminated by the passing of the trailing $b_3 = a/6[21\bar{1}]$ partial dislocation. Fig. 9(b) illustrates the stacking fault created on $(1\bar{1}1)$ in between the two dislocations. Apparently, a higher SFE leads to a lower “d” value (the width) of the stacking fault.

The above dislocation reaction and the partial dislocation movement are all restricted in a single $\{111\}$ plane. When the process is hindered in this plane, further deformation is difficult to proceed. Deformation may continue only by recombining the b_2 and b_3 partial dislocations back into the original full screw dislocation, which is able to bypass the obstacle by cross-slip. Given that b_2 and b_3 are 30° mixed dislocations, their edge ($b_{\perp}$) and screw ($b_{\parallel}$) components follow the relationships of $b_{(2)\parallel} = b_{(3)\parallel}$, $b_{(2)\perp} = -b_{(3)\perp}$ and $|b_{(2)\parallel}| > |b_{(2)\perp}|$. Based on these relationships, the interaction forces between the two partial dislocations can also be derived. The forces are attractive between the edge components, but repulsive between the screw components of the two dislocations. The magnitude of the repulsive force is larger than that of the attractive force, giving a net repulsive interaction between the two. In fact this is the essence of the separation of the two partial dislocations and the formation of the stacking fault. Thus, the recombination of the two partial dislocations is inherently prohibited.

It is also easy to understand that a larger stacking fault spacing implies a stronger repulsive interaction between the two partial dislocations, thus it requires more additional energy for them to overcome the resistance to recombine. This implies higher strength of the matrix. In other words, a decrease in stacking fault energy helps to strengthen the alloy.

Fig. 9(c) shows the evolution of the stacking fault formation

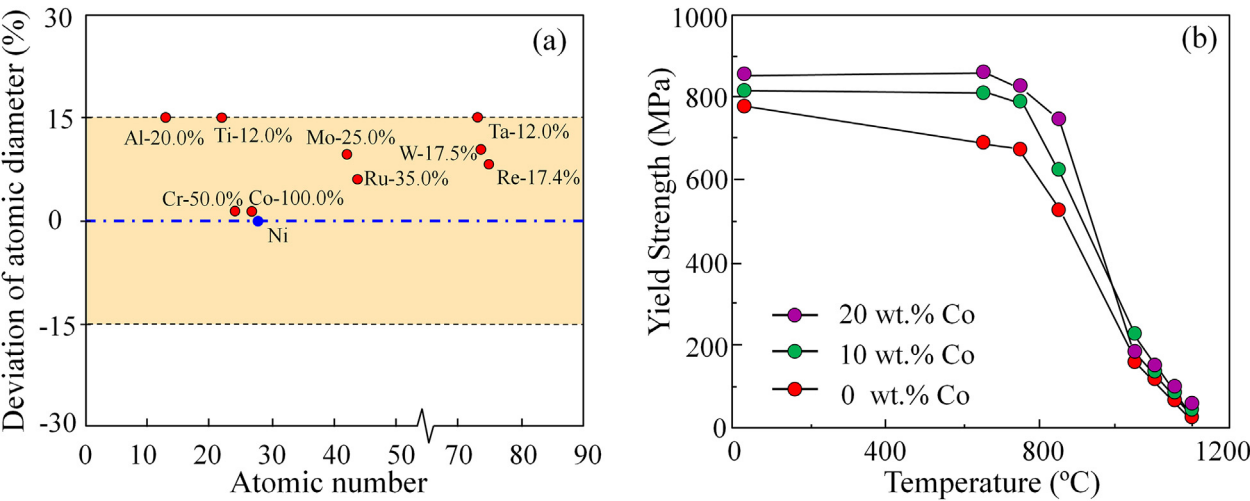


Fig. 8. Effects of solid strengthening elements in Ni-based superalloys. (a) Deviations of atomic diameters from Ni and the maximum solubilities of different solid solution elements in Ni-based superalloys [106]. (b) Effect of Co content on the yield strength of U720LI based alloy [107].

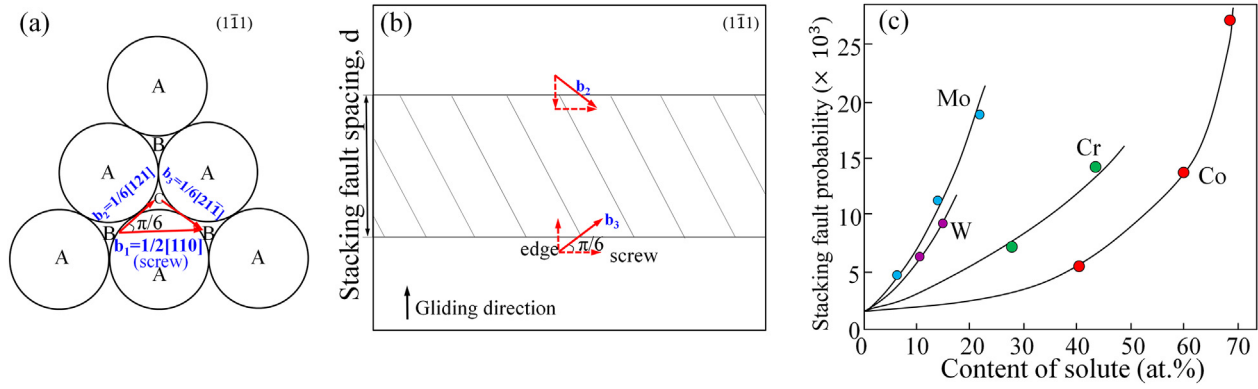


Fig. 9. Illustration of stacking faults in FCC crystal. (a) The close-packed (111) basal plane of FCC structure and the full and partial dislocations. (b) The dissociation of a screw dislocation into two Shockley partial dislocations and the stacking fault formed in between the partial dislocations. (c) Effect of alloying elements on the probability of occurrence of stacking faults in Ni-based alloys [109,110].

probability with solute concentration for a few solid solution elements [109,110]. It is seen that the probability increase with increasing the contents of all the four elements of Co, Cr, Mo and W. The stacking fault probability is inversely proportional to stacking fault energy. This observation implies that Co, Cr, Mo and W lower the stacking fault energy in Ni–Al system. Similar findings have

also been reported by other studies [110–114]. In addition, Re and Ru have also been found to exhibit the same effects in Ni–Al alloys [115,116].

3.3.3. Melting temperature

The alloying elements also influence the liquidus and solidus of

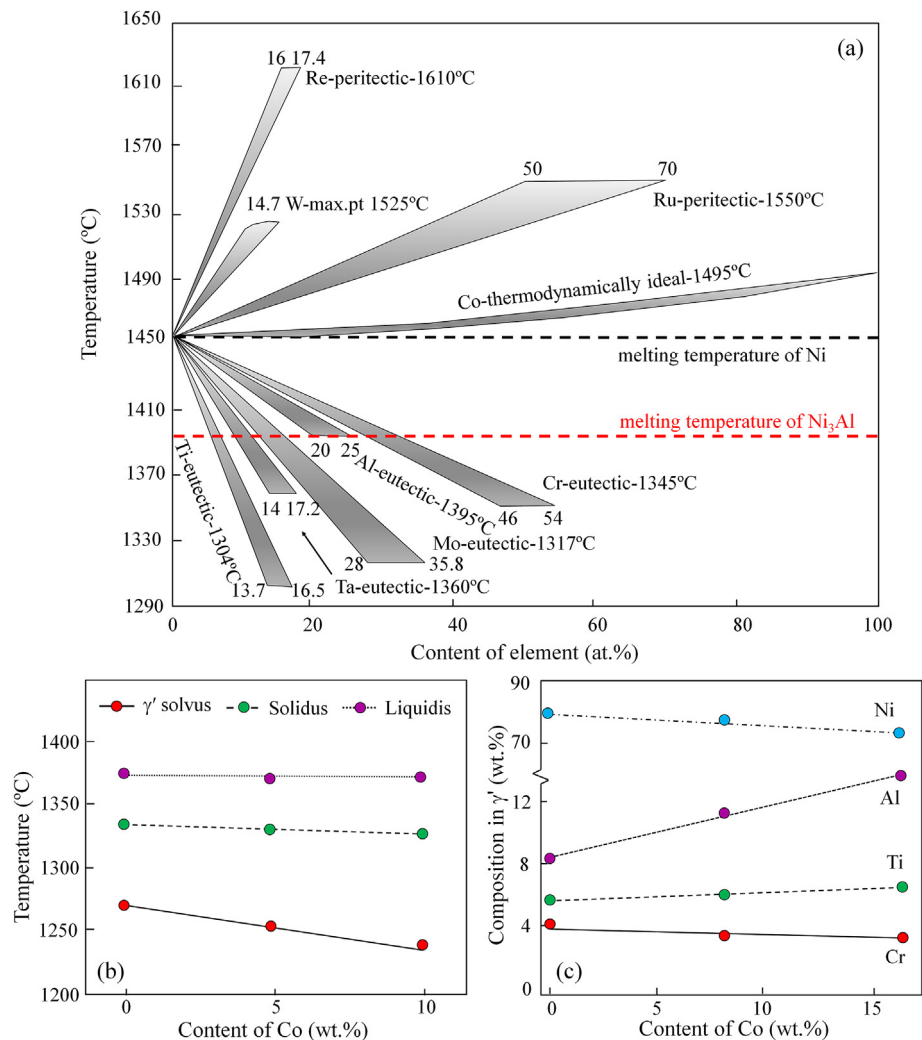


Fig. 10. The variation of solidus and liquidus lines with different content of elements. (a) The solidus and liquidus lines for the Ni-X binary alloys [10,117]. Variation of solution temperature (b) and compositions (c) in γ' phase as the function of content of Co [108].

Ni-based superalloys. Fig. 10(a) shows the liquidus and solidus lines of different Ni-X binary alloys. The figure is reconstructed based on the original figures and data reported in Refs. [10,117]. The horizon line represents the melting temperature of Ni (1450 °C). The elements can be classified into two types based on their effects on the liquidus and solidus, with Re, W, Ru and Co increasing the liquidus and solidus temperatures (referred to as type I) and Ti, Mo, Ta and Cr decreasing the liquidus and solidus temperatures (referred to as type II). Similar effects have also been observed in commercial Ni-based superalloys [99,118–123].

To ensure that the melting temperature of a superalloy is above that of the γ' phase (1395 °C as for Ni₃Al, which is indicated by the red line in Fig. 10(a)), addition of type II elements must be limited to below a suitable amount. Based on the data presented in Fig. 10(a), the maximum concentrations of the elements can be estimated to be ~5.3 at.% for Ti, ~8.5 at.% for Ta, ~12.3 at.% for Mo and ~26.5 at.% for Cr. As seen in Table 1, the maximum amounts of these elements in commercial Ni-based single crystalline superalloys are ~2.9 at.% for Ti, ~4.2 at.% for Ta, ~2.9 at.% for Mo and ~11.6 at.% for Cr. It is seen that these values are all above the actual contents of the elements used in commercial Ni-based superalloys (Table 1).

Fig. 10(b) shows the effects of Co content on the liquidus and solidus temperatures of the Mar-M247 superalloy and the solvus temperature of the γ' phase in the alloy [108]. Co has a weak influence on the liquidus and solidus temperatures of the alloy, as also expressed in Fig. 10(a), but a more obvious influence on the solvus temperature of the γ' phase, decreasing by ~30 °C with the addition of 10 at.% of Co. The different impacts of Co content on the liquidus of the alloy and the solvus of the γ' phase is attributed to the effect of element partition in the alloy.

Fig. 10(c) shows the effect of Co content on the composition of the γ' phase in Mar-M247 alloy. The addition of Co has negligible influence on the contents of Ni, Cr and Ti in the γ' phase but strongly increases the Al content. This is because that Co has a negative segregation coefficient ($k_{\gamma'}$) in the γ' phase, i.e., it is a γ phase formation element. Thus a higher Co content in the γ phase displaces and encourages γ' phase formation elements, such as Al and Ti (and Ta), to segregate more into the γ' phase.

3.3.4. Other effect

In addition to the effects discussed above, alloying elements also exhibit other effects to the alloy, e.g., interface segregation of W and Re, which retards coarsening rate of the γ' phase. Fig. 11(a) shows the diffusion coefficients of some common alloying elements in Ni, as a function of the inverse temperature [124]. Re and W are the elements with the lowest diffusivities among all the elements presented in Fig. 11(a). The low diffusivity of elements originates from their high numbers of unpaired d shell electrons, which provide high energy barriers for solute-vacancy exchange, which is essential for diffusion [10,125,126]. The different diffusion coefficients of the elements in Ni affect the microstructure stability of the superalloys, in particular, retarding the coarsening of the γ' phase and precipitation of the TCP phases.

Both Re and W have negative segregation coefficients in γ' , e.g., ~12.2 for Re and ~1.3 for W in a Ru-free superalloy [127]. During γ' phase coarsening (or γ' precipitation heat treatment for the same argument), in the region where the γ phase is to be converted to γ' , Re and W have to be expelled out via diffusion. Their low diffusivities make their diffusion slow and difficult and the piling-up of these elements at the γ'/γ interfaces during γ' phase growth is also slow to dissipate. This slows down the coarsening rate of the γ' phase [128–132]. Fig. 11(b)–(e) show the comparison of γ' sizes in four alloys after a 2-h aging treatment at 1065 °C [29]. These four alloys contained different contents of Re and W. The alloys with more Re (Fig. 11(c), (d) and (e)) exhibited smaller γ' cuboids than those without Re (Fig. 11(b)), demonstrating that addition of Re reduces the coarsening rate of the γ' phase. The γ' size in the Alloy 1444+6 (6 wt.%Re) is about 2/3 of that in Alloy 1444 (0 wt.%Re) after the heat treatment.

3.3.5. TCP phase formation

Addition of Cr [133–136], Mo [137–139], W [140,141] and Re [52,142] has been reported to promote the formation of TCP phases. Table 4 lists the structures and constituent elements of the TCP phases observed in Ni-based single crystalline superalloys [142]. Precipitation of TCP phases will consumes these elements and thus reduces their strengthening effect on the γ matrix. The TCP phase

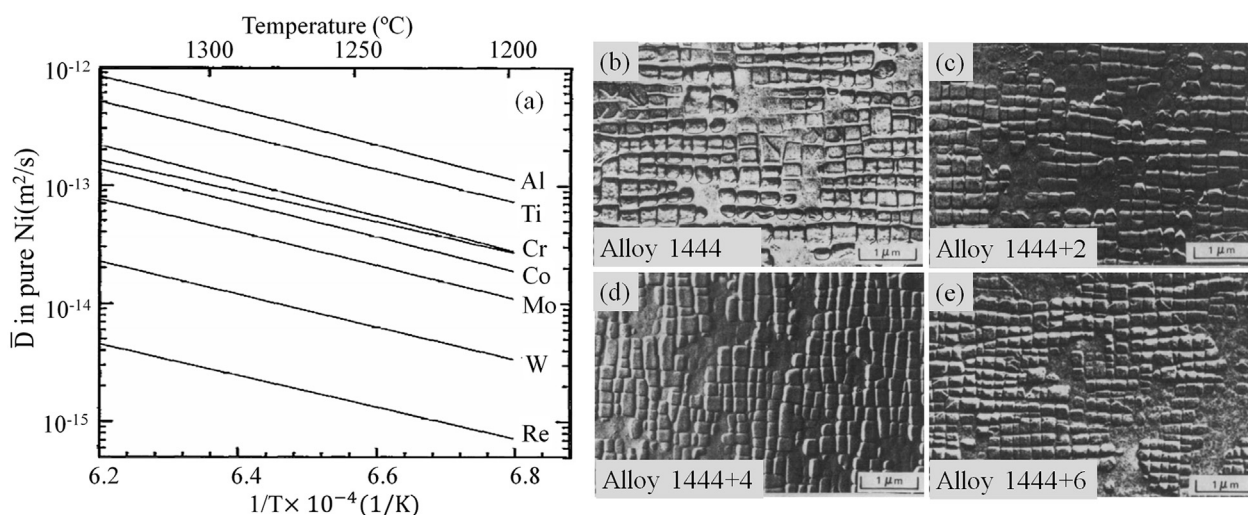


Fig. 11. Effect of element diffusivity on γ' phase coarsening of Ni-based superalloys. (a) Diffusion coefficients of different elements in Ni [124]. (b)–(e) Microstructure of four 1444 alloys containing different amount of Re. The samples were all aged for 2 h at 1065 °C. The original Alloy 1444 is shown in (b), the alloys shown in (c)–(e) are obtained by substituting W with 2, 4, 6 wt.% Re in the base of Alloy 1444, respectively [29].

Table 4

Common TCP phases in Ni-based single crystalline superalloy [142].

Type	Crystal structure	Space group	Atoms per Unite Cell	Lattice parameter (Å)	α
σ	Tetragonal	P42/mnm	30	a = b = 8.78; c = 4.54	90.0°
μ	Rhombohedra	R-3m	13	a = b = c = 9.04	30.5°
P	Hexagonal		39	a = b = 4.75; c = 25.83	120.0°
	Orthorhombic	Pnma	56	a = 16.98; b = 4.75; c = 9.07	90.0°
R	Rhombohedra	R-3m	159	a = b = 10.93; c = 19.34	120.0°

precipitates also act as nucleation sites of micro-cracks, thus are harmful for creep resistance of the superalloys [143–146].

3.4. Grain boundary elements

Grain boundary elements refer to C and B, which form carbide and boride particles along grain boundaries and cause strengthening by hindering the movement of grain boundaries in Ni-based superalloys. These elements were first removed in the first generation single crystalline superalloys, since grain-boundary strengthening is unnecessary [10]. It is then recognized that these elements help to reduce micro-segregation of elements and the volume fraction of eutectic component in the microstructure, leading to a reduction of casting pore and hot tearing [147–149]. These elements are then added back in the following generations of single crystalline superalloys [10]. C and B are found to react with other heavy elements in the alloys to form carbides and borides, and to strengthen low angle grain boundaries [150–152]. With these combined effects, the grain boundary elements prolong the service life of superalloys [150]. However, over addition of these elements can promote over growth of carbides and borides and induce their transformation to TCP phases, thus decrease the rupture life of superalloys [153–155]. In addition, C and B decrease the solidus temperature of the alloys, weakening their thermal stability [17,156,157].

Table 5 lists the structures and constituent elements of the carbides and borides observed in Ni-based single crystalline superalloys [1]. Formation of the carbides and borides consumes both the precipitation and solid strengthening elements. Therefore, the amount of the grain boundary elements is always controlled to be low, e.g., <0.05 wt.% for C and <0.004 wt.% for B, as shown in Table 1.

3.5. Lattice misfit

3.5.1. Effects of alloying elements on lattice misfit

The lattice misfit (δ) between the γ' and γ phases has strong influence on the density and evolution rate of dislocation networks and the creep to rupture life of superalloys [1]. The lattice misfit of superalloys can be adjusted by controlling the segregation of the alloying elements in the γ' and γ phases [138,158–163]. The lattice misfit is defined as [10]:

$$\delta = 2 \times (\alpha_{\gamma'} - \alpha_{\gamma}) / (\alpha_{\gamma'} + \alpha_{\gamma}) \quad (2-1)$$

where α_{γ} and $\alpha_{\gamma'}$ are lattice parameters of the γ and γ' phases, respectively.

The atomic diameter of all the alloying elements follow the order of Ni < Co < Cr < Ru < Re < Mo < W < Al < Ta < Ti, i.e., all the common alloying elements used in Ni-based superalloys are larger than Ni. Thus, addition of these elements into the Ni lattice always increases the lattice parameter of the γ phase. In comparison, the γ' phase has a L1₂ structure and contains both Ni and Al. The diameter of Al is larger than all of the alloying elements except Ta and Ti. Therefore, addition of the elements to the Ni site increases the lattice parameter of γ' phase whereas addition of the elements (except Ta and Ti) to the Al site decreases its lattice parameter. The lattice misfit between the γ and γ' phases is influenced by all the alloying elements. Empirical rules for estimating the lattice parameters of the γ and γ' phases are given in equations (2-2) and (2-3) [10].

$$\alpha_{\gamma} = 3.524 + 0.110x_{Cr}^{\gamma} + 0.478x_{Mo}^{\gamma} + 0.444x_{W}^{\gamma} + 0.441x_{Re}^{\gamma} + 0.179x_{Al}^{\gamma} + 0.422x_{Ti}^{\gamma} + 0.700x_{Ta}^{\gamma} \quad (2-2)$$

$$\alpha_{\gamma'} = 3.570 - 0.004x_{Cr}^{\gamma'} + 0.208x_{Mo}^{\gamma'} + 0.194x_{W}^{\gamma'} + 0.262x_{Re}^{\gamma'} + 0.258x_{Ti}^{\gamma'} + 0.500x_{Ta}^{\gamma'} \quad (2-3)$$

where x_i^{γ} and $x_i^{\gamma'}$ are the mole fractions of element i in γ and γ' , respectively.

3.5.2. Influence of lattice misfit on γ' morphology

As presented in Fig. 3(c), Ni-based superalloys having cuboidal shaped γ' phase exhibit the longest creep life. Thus, it is important to achieve cuboidal shaped γ' phase particles. Nathal suggested that cuboidal shaped γ' phase particles can only be obtained within a certain range of misfit values, and proposed the following general rule [75]:

$$0.1\% < |\delta| < 0.5\% \quad (2-4)$$

Fig. 12(a) shows the influence of misfit on the microstructures of three Ni-based superalloys, including USTB F1, F5 and F6 [164]. The lattice misfits of the three alloys are 0.04% for F1, -0.54% for F2 and -1.01% for F6. The morphologies of the γ' phase in the three alloys after precipitation heat treatment are spherical in F1, cuboidal in F2 and irregular raft in F6. The three alloys were subjected to thermal exposure under no stress at 1100 °C for different times. The γ' phase in the F1 alloy remained spherical up to 2000 h of thermal exposure. The γ' phase particles in the F2 alloy evolved from cuboidal to irregular raft in shape after 2000 h of exposure. The initial raft γ' phase in the F6 alloy coarsened gradually with time during the thermal exposure.

These observations demonstrate that in alloys with small absolute lattice misfit, the γ' phase will form in the spherical shape, obviously to have minimum interface area, thus the lowest interface energy [165]. For alloys with higher absolute lattice misfit, the γ' phase is formed in cuboidal shapes in order to minimize the elastic strain energy associated with the lattice distortion [166,167]. The Young's modulus of γ and γ' cubic phases is the lowest in the <100> directions, leading to the formation of the {100} faceted cuboidal shaped γ' phase particles. There is lack of clear explanation in the literature of the formation of irregular raft morphology of the

Table 5

Common carbides and borides in Ni-based single crystalline superalloy [1].

Phases	Structure	Parameter (Å)	A (°)	Constituent
MC	Cubic	a = 4.300	90.0	Ti, Ta, Hf
M ₆ C	Cubic	a = 1.085	90.0	Mo, W
M ₂₃ C ₆	Cubic	a = 10.500	90.0	Cr, Mo, W
M ₇ C ₃	Hexagonal	a = 13.980, c = 4.523	90.0	Cr
M ₃ B ₂	Tetragonal	a = b = 5.600, c = 3.300	90.0	Ta, Mo, Cr

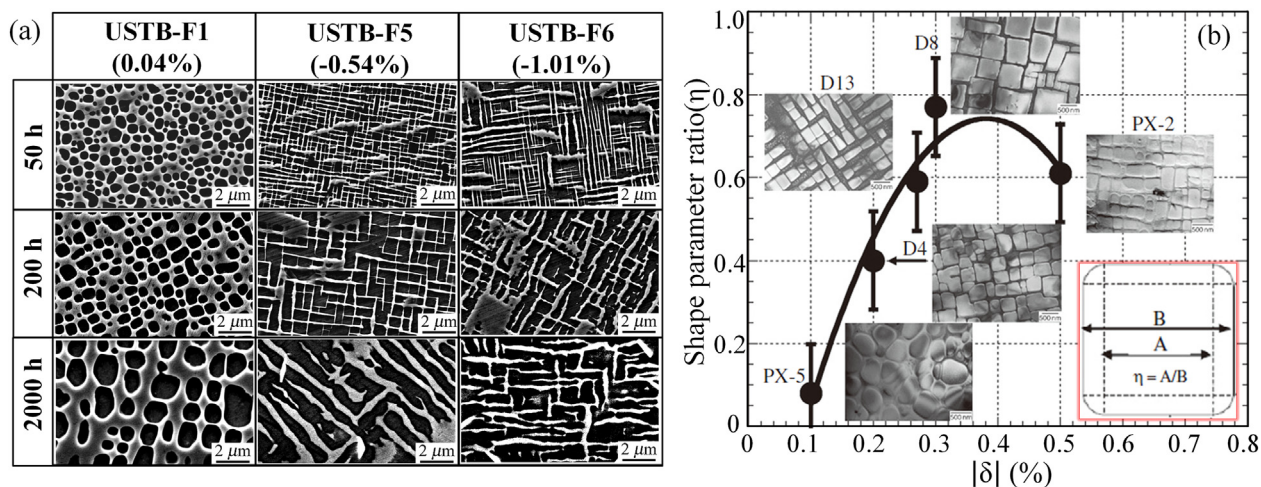


Fig. 12. Correlation between lattice misfit and morphology of γ' phase. (a) Typical microstructure of USTB F1, F5, and F6 alloys after exposing at 1100 °C for different times [164]. (b) Shape parameter ratio as a function of high temperature lattice misfit magnitude [172].

γ' phase, usually observed in alloys of excessively high misfit. It has been attributed to the combined effects of elasticity, energy and diffusion [168–171] without any specific explanation how each of these factors may lead to the particular structure. A possible reason is that with excessively high lattice misfit, the lattice distortion energy isotropy among the three principal directions are easily disturbed, and one accidental overgrowth in one direction will break the fragile balance and lead to rapid abnormal growth into e non-cubical shapes in random directions.

The shape of a γ' phase particle may be described in two aspects: the aspect ratio and cubicity factor. The aspect ratio (λ) may be defined as the ratio of the longest axis to the shortest axis of a particle: ($\lambda = L_{\text{long}}/L_{\text{short}}$). Obviously the value of λ is always ≥ 1 and that $\lambda = 1$ defines an equiaxial body and $\lambda > 1$ defines a disc or a rod.

An equiaxial particle with $\lambda = 1$ may also be a perfect cube or a sphere, and a γ' phase particle in a Ni-based superalloy is often a cuboid with rounded edges and corners, i.e. between the two ideal shapes. To describe the degree of cubicity (or sphericity), the shape parameter ratio (η) has been used. The η is defined as the ratio of the length of a cube with rounded edges and corners to the length

of the largest square edged cube left after cutting off the sides of the original cube to remove the rounded edges and corners, as illustrated in the inset in Fig. 12(b). Obviously $\eta = 0$ defines a perfect sphere and $\eta = 1$ defines a perfect cube.

Fig. 12(b) shows the dependence of the γ' phase morphology and the shape parameter ratio (η) on lattice misfit [172]. It is seen that the η increases with increasing $|\delta|$ to a maximum of ~ 0.75 at $|\delta| = 0.4\%$ and then decreases with further increase of $|\delta|$. This phenomenon was also observed in other commercial Ni-based superalloys and similar critical values of $|\delta|$ (0.4%) and η (0.75) were obtained [173–176].

The shape parameter ratio can be theoretically predicted to range from 0.1 to 0.8. The reason why it does not approach unity is because of the influence of the high elastic stresses and specific surface energies at the sharp edges and corners of a particle [172]. The maximum theoretical and experimental values of η are in close agreement. This implies that perfectly cubic γ' phase particles with $\eta = 1$ can hardly form.

Sluytman and Pollock [172] showed that the rupture life of superalloys increases with increasing η . Therefore, to achieve a

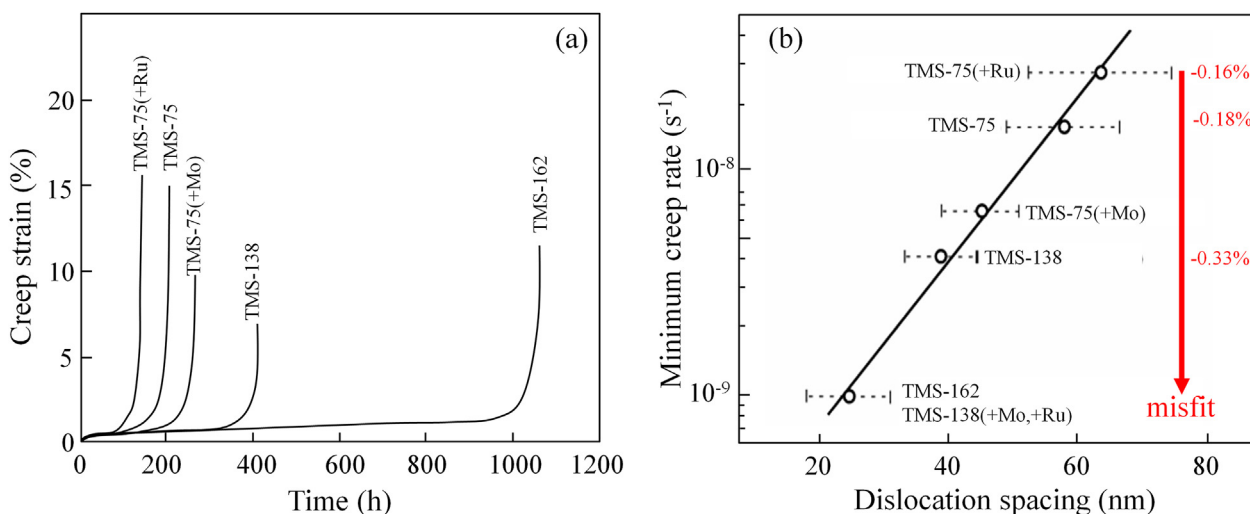


Fig. 13. Correlation between the lattice misfit and the minimum creep rate and the creep life. (a) Creep curves of five superalloys tested at 1100 °C/137 MPa [177]. (b) Minimum creep rates as a function of interfacial dislocation network spacing for the superalloys shown in Fig. 10(a) [178].

high rupture life the absolute misfit should be tuned to a value close to 0.4% for an as high η value as possible (Fig. 12(b)).

3.5.3. Influence of lattice misfit on dislocation network at the γ'/γ interface

Zhang et al. studied the influence of lattice misfit on dislocation networks formed at γ'/γ interfaces during creep testing [177,178]. Fig. 13(a) shows the creep curves of different Ni-based superalloys tested at 1100 °C under a bias tensile stress of 137 MPa. These alloys have different lattice misfit values. Upon creep testing, they all develop γ'/γ interface dislocation networks, but of different average cell spacing values. The rupture life of TMS-162, which has a very small dislocation spacing, is 1000 h, which is ~10 times of that of TMS-75, which has a very large dislocation spacing. A long creep life implies a low creep rate. Fig. 13(b) shows the correlation between the minimum creep rate of these superalloys and the interfacial dislocation network spacing. It is seen that creep rate decreases practically linearly with decreasing dislocation cell spacing in the network. This is because that dislocation networks of smaller spacings are more effective in blocking dislocation slip to the γ'/γ interface, due to the repulsive forces between like dislocations within the network, thus resulting in lower creep rates. Also displayed in the figure are the misfit values of the alloys. It is seen that smaller dislocation spacing values are correlated to higher absolute misfit values. It is known that dislocations are attracted to the γ'/γ interfaces to relax the misfit [179,180]. Thus, an interface of higher absolute misfit will attract and can accommodate more dislocations, thus resulting in lower dislocation network spacings [181,182].

The magnitude of lattice misfit in an alloy is most directly controlled by its chemical composition. For example, TMS-162 is designed based on TMS-138 by adding Mo and Ru to increase the magnitude of the negative lattice misfit of the alloy. Mo and Ru are both γ phase forming elements, and are larger atoms than Ni. Enrichment of Mo and Ru in the γ phase further expand the lattice parameter of the γ phase, thus enlarge the negative misfit of the alloy. This design brings the alloy misfit to be much closer to the optimal misfit of 0.4 for creep rupture life (section 3.5.2) [172]. As a result, alloy TMS-162 exhibits a reduced average dislocation spacing in the network and a prolonged creep rupture life than

alloy TMS-138 [183,184].

3.6. Long term stability elements

Ni-based superalloys undergo continuous microstructural changes during high temperature exposure. One such change is the formation of TCP phases during long time thermal exposure or creep testing. Formation of TCP phases lowers the strength of the alloys, by consuming the strengthening elements in the matrix and producing high internal stresses at the interfaces for crack nucleation. Thus it is of interest to suppress the formation of TCP phases in order to improve the long term microstructural stability of Ni-based superalloys. Ru has been found to be an effective element to inhibit the formation of TCP phases due to its “reverse partition effect” effect [127,185–189]. The “reverse partition effect” refers to the influence of an element in weakening or reversing the segregation behaviors of the γ and γ' phase forming elements in their respective phases.

Fig. 14(a) compares the partition ratios of different elements in two alloys with different Ru contents, denoted 0Ru and 3Ru [190]. In 0Ru alloy, Cr, Co, Mo and Re segregate to the γ phase whereas Al and Ta concentrate in the γ' phase. The addition of 3% Ru in 3Ru alloy reduced the partition ratios (weakens the segregation) of Co, Cr and Re and also Al and Ta. The underlying mechanism for this phenomenon is still unclear. However, a reversed segregation, or a lowered partition ratio, for an element implies an increased solubility and tolerance of the matrix from which the element was ejected, thus lowers the probability and driving force for the precipitation of TCP phases. Ru has also been found to increase the solubility of Re in the γ phase [186] and the misfit strain energy at the TCP/ γ matrix interfaces. The latter further lowers the driving force for TCP phase precipitation [191–193].

Fig. 14(b) plots the volume fractions (or area fraction as in 2D metallography) of TCP precipitates formed in three alloys of different Ru contents as functions of time and temperature [32]. The volume fraction of TCP precipitates increases with increasing temperature and time for the Ru-free (0Ru) alloy, demonstrating its thermal instability. The addition of Ru significantly reduced the precipitation of TCP phases.

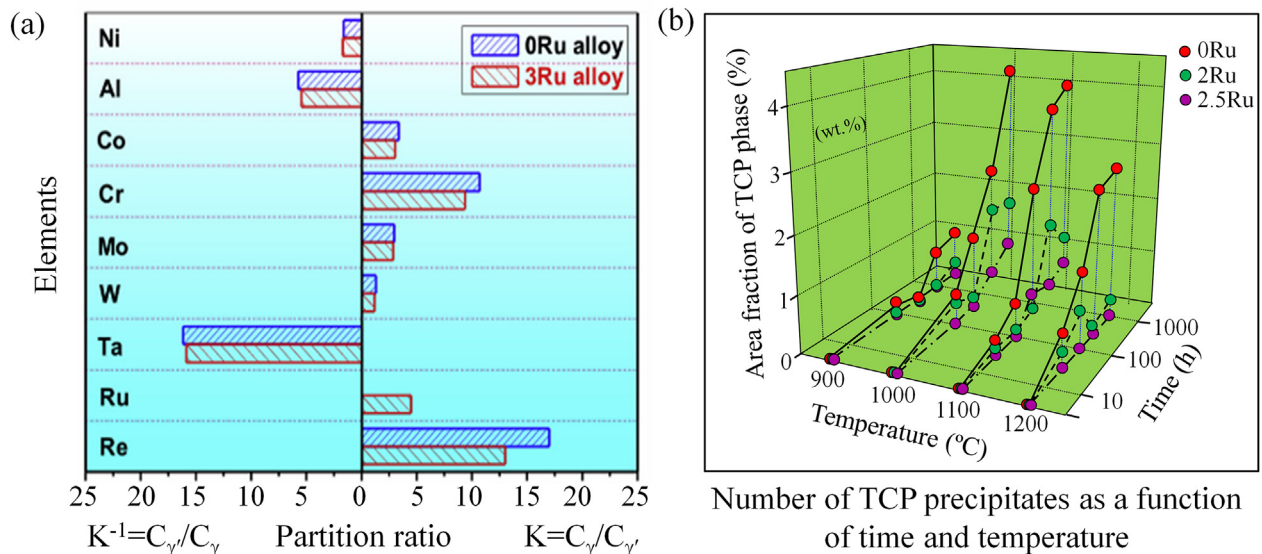


Fig. 14. (a) Partition behavior of different elements in the 0Ru and 3Ru alloy detected by 3 dimensional atom probe tomography [190]. (b) Number of TCP precipitates as a function of time and temperature [32].

4. Oxidation resistance elements

Ni-based superalloys are generally oxidation resistant. In service, however, they rely on protective thermal barrier coatings (TBC) to be their first defense against oxidation at elevated temperature [194,195]. Thermal barrier coatings usually contain two layers. The out layer is made of a ceramic which insulates the alloy from direct exposure to high temperature. The inner layer is a metal based coating to produce an in-situ protective Al_2O_3 layer [196,197]. They are applied to the superalloys using either of two common methods, i.e., plasma spraying or electron-beam physical vapor deposition [198–200]. Given the ceramic compound nature of the thermal barrier coatings, they have significantly different thermal expansion coefficients from that of Ni-based superalloys. This causes the thermal barrier coatings to fatigue by thermal stresses and to crack and fall off the metal surface. This is the main drawback of thermal barrier coatings [201–203]. This also requires Ni-based superalloys to have adequate intrinsic oxidation resistance.

Fig. 15(a) shows the Ellingham diagram of formation Gibbs free energies of three common oxides in Ni-based superalloys [26]. The formation Gibbs free energies of the oxides follow the order of $G_{\text{Al}_2\text{O}_3} < G_{\text{Cr}_2\text{O}_3} < G_{\text{NiO}}$, indicating their relative thermal stabilities. NiO is an oxide of high intrinsic concentration of vacancies and thus forms rapidly with voids. Cr_2O_3 will not form a protective surface layer when Cr content is below ~12 wt.%. Ni-based superalloys modestly derive their oxidation resistance from the stable Al_2O_3 oxide. It has been confirmed experimentally that the oxidation rate of superalloys is low when Al_2O_3 forms [204–207]. Al is added in Ni-based superalloys usually at a concentration of ~6 wt.%, largely for the purpose of guarantee the formation of Al_2O_3 protective oxide [13, 221, 223, 226]. Cr is also added for the same effect, usually to below ~12 wt.% [10,205,208]. Fig. 15(b) shows the oxidation metal losses of PWA1480 (5.0 wt.%Al) and PWA1484 (5.6 wt.%Al) single crystalline superalloys [209]. It is seen that increasing Al content from 5.0 wt.% to 5.6 wt.% led to a significant reduction of oxidation rate. The service life of PWA1484 is also found to be 60% longer than that of PWA 1480 at 1050 °C [209].

5. Summary

The containing elements in single crystalline superalloy can major divide into different types. The Base elements, mechanical strengthening elements, long term stability elements and the oxidation resistance elements. The mechanical strengthening elements can further divide into precipitation, solid solution and the grain boundary elements. The major characters of them are listed as follow:

1. The precipitation elements consist of the Al, Ti and Ta. The Al element is to form the precipitation strengthening phase γ' with the Ni_3Al structure. It is believed the maximum service life can obtain with the volume fraction of γ' to be 60%–75%, and in cubic shape with the initial size of 0.2–0.5 μm . The Ti and Ta elements can increase the APB in γ' which in further will improve the ordered strengthening effect, and the peak value established when Ti/Al and Ta/Al value reach ~1 and ~0.2 respectively.
2. The solid solution elements consist of Cr, Co, Mo, W, Re and Ru. They all contain a higher atomic diameter than Ni, thus the solid effect can realized by lattice distortion. They also decrease the SFE of the alloy, which can inhibit the dislocation to cross slip. The Re, W, Ru and Co elements can increase the solidus temperatures of the alloy, while decreases with the Al, Ta, Ti Mo and Cr elements. These determined the thermal stability of the phases. Besides, the low diffusivity of W and Re elements make them segregated at the γ'/γ interface, which can in further inhibit the coarsen of γ' phase. The over adding of solid strengthening elements can promote the precipitation of TCP phase, which can dramatically decrease the creep life.
3. The grain boundary elements are consisting of C, B, and Hf. They can decrease the casting pore and hot tearing behavior, which also can strengthen the low angle boundary, these two behavior all perform in the manners of forming carbides and borides. But over adding the grain boundary elements can promote the growth of carbides and borides, which in further will cost both the precipitation and solid strengthening elements.

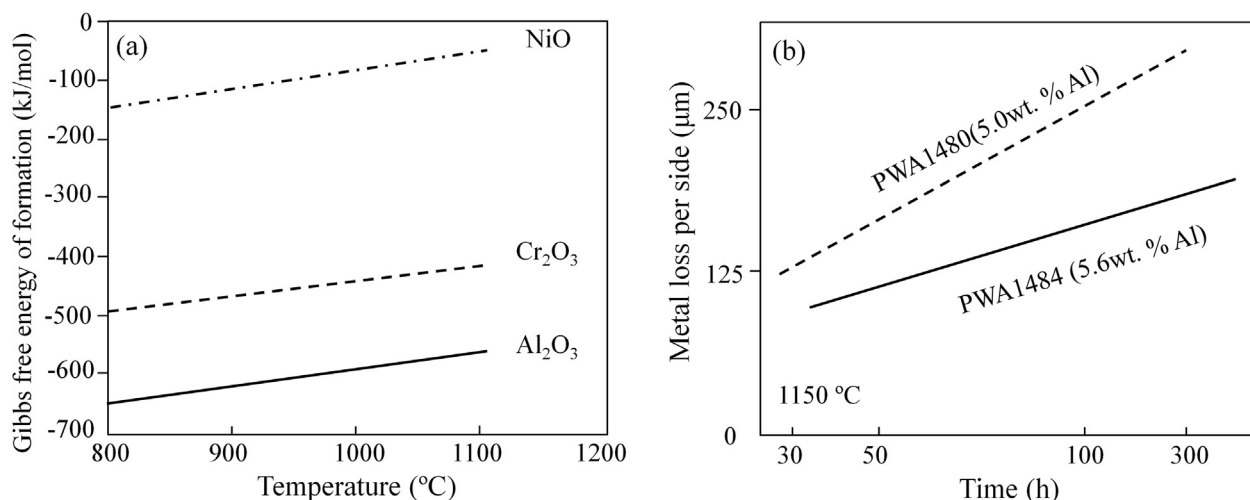


Fig. 15. Stability of oxidation and its influence on metal loss rate. (a) Ellingham diagram of Ni-based single crystalline superalloy [26]. (b) The rate of metal loss causing by oxidation in the PWA1480 and PWA1484 single-crystal superalloys [209].

4. The different proportion of precipitation and solid strengthening elements can adjust the misfit at the γ'/γ interface, which in further can influence the morphology of γ' and the dislocation network at the γ'/γ interface. It is found when the misfit reach -0.4% , a more cubic γ' is established, so as the denser dislocation network at the γ'/γ interface, these are all benefit for the service life.
5. The Ru indicates to be the long term stability element. It can change the partition behavior of other elements, named as “reverse partition” behavior. This behavior make more consisting elements of TCP phase enter into γ' phase, and thus inhibit the precipitation of TCP phase.
6. The alumina elements indicates to have the slowest oxidation rate. To ensure the formation of it, the content of Al should around $\sim 6\text{ wt.}\%$ and the Cr content should below $\sim 12\text{ wt.}\%$.

Acknowledgement

This work was supported by Grants from National Key R&D Program of China (2017YFB0305501), NSFC (51471008, 11327901) and the Importation and Development of High-Caliber Talents Project of Beijing Municipal Institutions (CIT&TCD201504013).

References

- [1] M.J. Donachie, S.J. Donachie, *Superalloys: a Technical Guide*, second ed., ASM International, 2002.
- [2] P. Auburtin, T. Wang, S.L. Cockcroft, A. Mitchell, Freckle formation in superalloys, *Metall. Mater. Trans. B* 31 (2000) 801–811.
- [3] H. Fecht, D. Furrer, Processing of nickel-base superalloys for turbine engine disc applications, *Adv. Eng. Mater.* 2 (2000) 777–787.
- [4] P. Caron, T. Khan, Evolution of Ni-based superalloys for single crystalline gas turbine blade applications, *Aerosol. Sci. Technol.* 3 (1999) 513–523.
- [5] M. Duhl, A.F. Giamei, A.F. Giamei, The development of single crystal superalloy turbine blades, *Superalloys 1980* (1980) 205–214.
- [6] R. Singer, Advanced materials and processes for land-based gas turbines, *Mater. Adv. Power Eng.* 2 (1994) 1707–1729.
- [7] N. Das, Advances in nickel-based cast superalloys, *Trans. Indian Inst. Met.* 63 (2010) 265–274.
- [8] A. Sato, J.J. Moverare, M. Hasselqvist, R.C. Reed, On the mechanical behavior of a new single-crystal superalloy for industrial gas turbine applications, *Metall. Mater. Trans.* 43 (2012) 2302–2315.
- [9] B. Geddes, H. Leon, X. Huang, *Superalloys Alloying and Performance*, ASM International, 2010.
- [10] R.C. Reed, *The Superalloys Fundamentals and Applications*, Cambridge University, Cambridge, 2006.
- [11] A. Devaux, E. Georges, P. Héritier, Development of new C&W superalloys for high temperature disk applications, *Adv. Mater. Res.* (2011) 405–410.
- [12] J.S. Benjamin, Dispersion strengthened superalloys by mechanical alloying, *Metall. Trans.* 1 (1970) 2943–2951.
- [13] M. McLean, Nickel-based alloys: recent developments for the aero-gas turbine, in: H.M. Flower (Ed.), *High Performance Materials in Aerospace*, Chapman & Hall, London, 1995, pp. 135–154.
- [14] W. Betteridge, S.W.S. Shaw, Development of superalloys, *Mater. Sci. Technol.* 3 (1987) 682–694.
- [15] T.M. Pollock, S. Tin, Nickel-based superalloys for advanced turbine engines: chemistry, microstructure and properties, *J. Propul. Power* 22 (2006) 361–374.
- [16] R.N. Jarrett, J.K. Tien, Effects of cobalt on structure, microchemistry and properties of a wrought nickel-base superalloy, *Mater. Trans.* (1982) 1021–1032.
- [17] R.V. Miner, Effects of C and Hf concentration on phase relations and microstructure of a wrought powder-metallurgy superalloy, *Metall. Mater. Trans.* (1977) 259–263.
- [18] J.K. Tien, J.C. Borofka, M.E. Casey, Precision cast vs. Wrought superalloys, *JOM* 38 (1986) 13–17.
- [19] E.O. Ezugwua, Z.M. Wanga, A.R. Machadob, The machinability of nickel-based alloys: a review, *J. Mater. Process. Technol.* 86 (1999) 1–16.
- [20] H. Saari, J. Beddoes, D.Y. Seo, L. Zhao, Development of directionally solidified γ -TiAl structures, *Intermetallics* 13 (2005) 937–943.
- [21] G.L. Erickson, K. Harris, DS and SX superalloys for industrial gas turbines, in: D. Coutouradis, et al. (Eds.), *Material for Advanced Power Engineering*, 1994, pp. 1055–1074.
- [22] A.J. Elliott, S. Tin, W.T. King, S.C. Huang, M.F.X. Gigliotti, T.M. Pollock, Directional solidification of large superalloy castings with radiation and liquid metal cooling (LMC): a comparative assessment, *Metall. Mater. Trans.* 35 (2004) 3221–3231.
- [23] N. D'Souza, M.G. Ardakani, M. McLean, Directional and single-crystal solidification of Ni-base superalloys: Part I. The role of curved isotherms on grain selection, *Metall. Mater. Trans.* 31 (2000) 2877–2886.
- [24] P. Caron, T. Khan, Improvement of Creep strength in a nickel-base single-crystal superalloy by heat treatment, *Mater. Sci. Eng.* 61 (1983) 173–184.
- [25] F.I. Versnyder, M.E. Shank, The development of columnar grain and single crystal high temperature materials through directional solidification, *Mater. Sci. Eng.* 6 (1970) 213–247.
- [26] A. Sato, Y.L. Chiu, R.C. Reed, Oxidation of nickel-based single-crystal superalloys for industrial gas turbine applications, *Acta Mater.* 59 (2011) 225–240.
- [27] K.E. Yoon, R.D. Noebe, D.N. Seidman, Effects of rhenium addition on the temporal evolution of the nanostructure and chemistry of a model Ni–Cr–Al superalloy. I: experimental observations, *Acta Mater.* 55 (2007) 1145–1157.
- [28] A.C. Yeh, S. Tin, Effects of Ru on the high-temperature phase stability of Ni-base single-crystal superalloys, *Metall. Mater. Trans.* 37 (2006) 2621–2631.
- [29] A.F. Giamei, D.L. Anton, Rhenium additions to a Ni-base superalloy: effects on microstructure, *Metall. Mater. Trans.* 16 (1985) 1997–2005.
- [30] Q. Wu, S. Li, Alloying element additions to Ni3Al: site preferences and effects on elastic properties from first-principles calculations, *Comput. Mater. Sci.* 53 (2012) 436–443.
- [31] R.C. Reed, T. Tao, N. Warnken, *Alloys-By-Design: application to nickel-based single crystal superalloys*, *Acta Mater.* 57 (2009) 5898–5913.
- [32] A. Sato, H. Harada, T. Yokokawa, T. Murakumo, The effects of ruthenium on the phase stability of fourth generation Ni-base single crystal superalloys, *Scr. Mater.* 54 (2006) 1679–1684.
- [33] K. Kawagishi, A. Sato, T. Kobayashi, H. Harada, Effect of alloying elements on the oxidation resistance of 4th generation Ni-base single-crystal superalloys, *J. Jpn. Inst. Metals* 69 (2005) 249–252.
- [34] C.L. Fu, R. Reed, A. Janotti, M. Krcmar, On the diffusion of alloying elements in the nickel-base superalloys, *Superalloys 2004* (2004) 867–876.
- [35] G.H. Meier, F.S. Pettit, J.L. Smialek, The effects of reactive element additions and sulfur removal on the adherence of alumina to Ni- and Fe-base alloys, *Mater. Corros.* 46 (1995) 232–240.
- [36] M. Morinaga, N. Yukawa, H. Adachi, Alloying effect on the electronic structure of Ni3Al (γ'), *J. Phys. Soc. Jpn.* 53 (1984) 653–663.
- [37] G.W. Meetham, Trace elements in superalloys-An overview, *Mater. Sci. Technol.* 11 (1984) 414–418.
- [38] J.S. Benjamin, Dispersion strengthened superalloys by mechanical alloying, *Metall. Mater. Trans.* B 1 (1970) 2943–2951.
- [39] D. Blavette, P. Caron, T. Khan, An atom probe investigation of the role of rhenium additions in improving creep resistance of Ni-base superalloys, *Scr. Mater.* 20 (1986) 1395–1400.
- [40] W.W. Milligan, S.D. Antolovich, Yielding and deformation behavior of the single crystal superalloy PWA 1480, *Metall. Mater. Trans.* 18 (1987) 85–95.
- [41] H. Rouault-Rogez, M. Dupeux, M. Ignat, High temperature tensile creep of CMSX-2 Nickel base superalloy single crystals, *Acta Mater.* 42 (1994) 3137–3148.
- [42] A. Sato, H. Harada, A.C. Yeh, K. Kawagishi, T. Kobayashi, A 5th generation SC superalloy with balanced high temperature properties and processability, *Superalloys 2008* (2008) 131–138.
- [43] Y. Takebe, T. Yokokawa, T. Kobayashi, K. Kawagishi, H. Harada, C. Masuda, Phase stability of nickel-base single crystal superalloys containing iridium substituting for ruthenium, *J. Jpn. Inst. Met.* 79 (2015) 227–231.
- [44] J.X. Zhang, H. Harada, Y. Ro, Y. Koizumi, T. Kobayashi, Thermomechanical fatigue mechanism in a modern single crystal nickel base superalloy TMS-82, *Acta Mater.* 56 (2008) 2975–2987.
- [45] K. Kawagishi, A.C. Yeh, T. Yokokawa, Development of an oxidation-resistant high-strength sixth-generation single-crystal superalloy TMS-238, *Superalloys 2012* (2012) 189–195.
- [46] G.L. Erickson, The development of the CMSX-11B and CMSX-11C alloys for industrial gas turbine application, *Superalloys 1996* (1996) 45–52.
- [47] M. Huang, J. Zhu, An overview of rhenium effect in single-crystal superalloys, *Rare Met.* 35 (2016) 127–139.
- [48] A. Volek, F. Pyczak, R.F. Singer, H. Mughrabi, Partitioning of Re between γ and γ' phase in nickel-base superalloys, *Scr. Mater.* 52 (2005) 141–145.
- [49] J. Yu, X. Sun, T. Jin, N. Zhao, H. Guan, Z. Hu, Effect of Re on deformation and slip systems of a Ni base single-crystal superalloy, *Mater. Sci. Eng.* 458 (2007) 39–43.
- [50] F. Pyczak, B. Devrient, F.C. Neuner, H. Mughrabi, A. Volek, The influence of different alloying elements on the development of the γ/γ' microstructure of nickel-base superalloys during high-temperature annealing and deformation, *Acta Mater.* 53 (2005) 3879–3891.
- [51] C.M.F. Rae, R.C. Reed, The precipitation of topologically close-packed phases in rhenium-containing superalloys, *Acta Mater.* 49 (2001) 4113–4125.
- [52] T.M. Pollock, The growth and elevated temperature stability of high refractory nickel-base single crystals, *Mater. Sci. Eng. B* 32 (1995) 255–266.
- [53] S. Xiang, S. Mao, H. Wei, Y. Liu, J. Zhang, Z. Shen, H. Long, H. Zhang, X. Wang, Z. Zhang, X. Han, Selective evolution of secondary γ' precipitation in a Ni-based single crystal superalloy both in the γ matrix and at the dislocation nodes, *Acta Mater.* 116 (2016) 343–353.
- [54] Y. Luo, Y. Zhao, S. Yang, J. Zhang, D. Tang, Effects of Ru on microstructure and stress rupture property of Ni-based single crystal superalloy DD22, *J. Aeronaut. Mater.* 36 (2016) 132–140.

- [55] E.V. Kozlov, E.L. Nikonenko, N.A. Koneva, Energy of planar defects in the Ni₃Al phase: theory and experiment, *Bull. Russ. Acad. Sci. Phys.* 71 (2007) 198–202.
- [56] Y. Wang, Z.K. Liu, L.Q. Chen, Thermodynamic properties of Al, Ni, NiAl, and Ni₃Al from first-principles calculations, *Acta Mater.* 52 (2004) 2665–2671.
- [57] P.A.J. Bagot, O.B.W. Silk, J.O. Douglas, S. Pedrazzini, D.J. Crudden, T.L. Martin, M.C. Hardy, M.P. Moody, R.C. Reed, An Atom Probe Tomography study of site preference and partitioning in a nickel-based superalloy, *Acta Mater.* 125 (2017) 156–165.
- [58] B.H. Ge, Y.S. Luo, J.R. Li, J. Zhu, Distribution of rhenium in a single crystal nickel-based superalloy, *Scr. Mater.* 63 (2010) 969–972.
- [59] Y. Xu, L. Zhang, J. Li, X. Xiao, X. Cao, G. Jia, Z. Shen, Relationship between Ti/Al ratio and stress-rupture properties in nickel-based superalloy, *Mater. Sci. Eng.* 544 (2012) 48–53.
- [60] H.S. Lee, D.H. Kim, D.S. Kim, K.B. Yoo, Microstructural changes by heat treatment for single crystal superalloy exposed at high temperature, *J. Alloy. Comp.* 561 (2013) 135–141.
- [61] V.A. Vorontsov, C. Shen, Y. Wang, D. Dye, C.M.F. Rae, Shearing of γ' precipitates by $a<112>$ dislocation ribbons in Ni-base superalloys: a phase field approach, *Acta Mater.* 58 (2010) 4110–4119.
- [62] A. Picasso, A. Somoza, A. Tolley, Nucleation, growth and coarsening of γ' -precipitates in a Ni–Cr–Al-based commercial superalloy during artificial aging, *J. Alloy. Comp.* 479 (2009) 129–133.
- [63] J.X. Zhang, H. Harada, Y. Koizumi, T. Kobayashi, Dislocation motion in the early stages of high-temperature low-stress creep in a single-crystal superalloy with a small lattice misfit, *J. Mater. Sci.* 45 (2009) 523–532.
- [64] J.X. Zhang, J.C. Wang, H. Harada, Y. Koizumi, The effect of lattice misfit on the dislocation motion in superalloys during high-temperature low-stress creep, *Acta Mater.* 53 (2005) 4623–4633.
- [65] M. Chandran, S.K. Sondhi, First-principle calculation of APB energy in Ni-based binary and ternary alloys, *Model. Simulat. Mater. Sci. Eng.* 19 (2011) 1–7, 025008.
- [66] C. Booth-Morrison, Z. Mao, R.D. Noebe, Chromium and tantalum site substitution patterns in Ni₃Al(L₁₂) γ' -precipitates, *Appl. Phys. Lett.* 93 (2008) 1–3, 033103.
- [67] C. Jiang, D.J. Sordelet, B. Gleeson, Site preference of ternary alloying elements in Ni₃Al: a first-principles study, *Acta Mater.* 54 (2006) 1147–1154.
- [68] A.V. Ruban, H.L. Skriver, Calculated site substitution in γ' -Ni₃Al, *Solid State Commun.* 99 (1996) 813–817.
- [69] T.M. Pollock, R.D. Field, Dislocations and high-temperature plastic deformation of superalloy single crystals, in: F.R.N. Nabarro, M.S. Duesbery (Eds.), *Dislocations in Solids*, vol. 11, 2002, pp. 547–618. Amsterdam.
- [70] M.V. Nathal, L.J. Ebert, The influence of cobalt, tantalum, and tungsten on the microstructure of single crystal nickel-base superalloys, *Metall. Mater. Trans.* 16 (1985) 1849–1862.
- [71] D.M. Shah, Orientation dependence of creep behavior of single crystal γ' (Ni₃Al), *Scr. Mater.* 17 (1983) 997–1002.
- [72] W.R. Johnson, C.R. Barrett, W.D. Nix, The high-temperature creep behavior of nickel-rich Ni–W solid solutions, *Metall. Mater. Trans. B* 3 (1972) 963–969.
- [73] F.R.N. Nabarro, M.S. Duesbery, *Dislocations in Solids*, vol. 11, Elsevier, Amsterdam, 2002.
- [74] T. Murakumo, T. Kobayashi, Y. Koizumi, H. Harada, Creep behaviour of Ni-base single-crystal superalloys with various γ' volume fraction, *Acta Mater.* 52 (2004) 3737–3744.
- [75] M.V. Nathal, Effect of initial gamma prime size on the elevated temperature creep properties of single crystal nickel base superalloys, *Metall. Mater. Trans.* 18 (1987) 1961–1970.
- [76] L.Z. He, Q. Zheng, X.F. Sun, H.R. Guan, Z.Q. Hu, High temperature low cycle fatigue behavior of Ni-base superalloy M963, *Mater. Sci. Eng.* 402 (2005) 33–41.
- [77] B.C. Wilson, J.A. Hickman, G.E. Fuchs, The effect of solution heat treatment on a single-crystal Ni-based superalloy, *JOM* 55 (2003) 35–40.
- [78] M.S.A. Karunaratne, D.C. Cox, P. Carter, R.C. Reed, Modelling of the micro-segregation in CMSX-4 superalloy and its homogenisation during heat treatment, *Superalloys 2000* (2000) 263–272.
- [79] P. Caron, T. Khan, Improvement of creep strength in a nickel-base single-crystal superalloy by heat treatment, *Mater. Sci. Eng.* 61 (1983) 173–184.
- [80] A. Pasturel, C. Colinet, A.T. Paxton, First-principles determination of the Ni–Al phase diagram, *J. Phys. Condens. Matter* 4 (1992) 945–959.
- [81] K. Morsi, Review: reaction synthesis processing of Ni–Al intermetallic materials, *Mater. Sci. Eng.* 299 (2001) 1–15.
- [82] F.J. Bremer, M. Beyss, E. Karthaus, A. Hellwig, Experimental analysis of the Ni–Al phase diagram, *J. Cryst. Growth* 87 (1988) 185–192.
- [83] Y. Tang, M. Huang, J. Xiong, J. Li, J. Zhu, Evolution of superdislocation structures during tertiary creep of a nickel-based single-crystal superalloy at high temperature and low stress, *Acta Mater.* 126 (2017) 336–345.
- [84] H. Long, Y. Liu, D. Kong, H. Wei, Y. Chen, S. Mao, Shearing mechanisms of stacking fault and anti-phase-boundary forming dislocation pairs in the γ' phase in Ni-based single crystal superalloy, *J. Alloy. Comp.* 724 (2017) 287–295.
- [85] D. Raynor, J.M. Silcock, Strengthening mechanisms in γ' precipitating alloys, *Met. Sci.* 4 (1970) 121–130.
- [86] F.R.N. Nabarro, D.F. Villiers, *Physics of Creep and Creep-resistant Alloys*, Taylor and Francis, 1995.
- [87] L.R. Liu, T. Jin, H.J. Chen, X.F. Sun, H.R. Guan, Z.Q. Hu, Effect of Ti/Al ratio on the microstructure and stress rupture property in a Ni-Base single crystal superalloy, *Rare Metal Mater. Eng.* 38 (2009) 612–616.
- [88] M.L. Tan, C.S. Wang, Y.A. Guo, J.T. Guo, L.Z. Zhou, Influence of Ti/Al ratios on γ' coarsening behavior and tensile properties of GH984G alloy during long-term thermal exposure, *Acta Metall. Sin.* 50 (2014) 1260–1268.
- [89] Y. Dayun, Z. Xuan, J. Tao, S. Xiaofeng, The influence of cobalt, tungsten, and titanium on stress-rupture properties of nickel-base single crystal superalloy, *Rare Metal Mater. Eng.* 34 (2005) 1295–1298.
- [90] J. Jones, D.J.C. Mackay, Neural network modelling of the mechanical properties of nickel base superalloys, *Superalloys 1996* (1996) 417–424.
- [91] V.A. Phillips, Coherency hardening in Ni–Al–Ti alloys, *Scr. Mater.* 2 (1968) 147–151.
- [92] Y. Nunomura, Y. Kaneno, H. Tsuda, T. Takasugi, Phase relation and microstructure in multi-phase intermetallic alloys based on Ni₃Al–Ni₃Ti–Ni₃V pseudo-ternary alloy system, *Intermetallics* 12 (2004) 389–399.
- [93] Y. Xu, C. Yang, X. Xiao, X. Cao, G. Jia, Z. Shen, Evolution of microstructure and mechanical properties of Ti modified superalloy Nimonic 80A, *Mater. Sci. Eng.* 530 (2011) 315–326.
- [94] J.P. Collier, P.W. Keefe, J.K. Tien, The effects of replacing the refractory elements W, Nb, and Ta with Mo in nickel-base superalloys on microstructural, microchemistry, and mechanical properties, *Metall. Mater. Trans.* 17 (1986) 651–661.
- [95] C. Booth-Morrison, R.D. Noebe, D.N. Seidman, Effects of tantalum on the temporal evolution of a model Ni–Al–Cr superalloy during phase decomposition, *Acta Mater.* 57 (2009) 909–920.
- [96] F. Xue, H.J. Zhou, X.F. Ding, M.L. Wang, Q. Feng, Improved high temperature γ' stability of Co–Al–W-base alloys containing Ti and Ta, *Mater. Lett.* 112 (2013) 215–218.
- [97] M. Ooshima, K. Tanaka, N.L. Okamoto, K. Kishida, Effects of quaternary alloying elements on the γ' solvus temperature of Co–Al–W based alloys with fcc/L₁₂ two-phase microstructures, *J. Alloy. Comp.* 508 (2010) 71–78.
- [98] E.C. Caldwell, F.J. Fela, G.E. Fuchs, The segregation of elements in high-refractory-content single-crystal nickel-based superalloys, *JOM* 56 (2004) 44–48.
- [99] S. Wöllmer, T. Mack, U. Glatzel, Influence of tungsten and rhenium concentration on creep properties of a second generation superalloy, *Mater. Sci. Eng.* 319–321 (2001) 792–795.
- [100] L.J. Carroll, Q. Feng, J.F. Mansfield, T.M. Pollock, High refractory, low misfit Ru-containing single-crystal superalloys, *Metall. Mater. Trans.* 37 (2006) 2927–2938.
- [101] F. Pyczak, B. Devrient, H. Mughrabi, The effects of different alloying elements on the thermal expansion coefficients, lattice constants and misfit of nickel-based superalloys investigated by x-ray diffraction, *Superalloys 2004* (2004) 827–836.
- [102] A.C. Yeh, S. Tin, Effects of Ru and Re additions on the high temperature flow stresses of Ni-base single crystal superalloys, *Scr. Mater.* 52 (2005) 519–524.
- [103] W.D. Callister, D.G. Rethwisch, *Fundamentals of Materials Science and Engineering: an Integrated Approach*, second ed., John Wiley and Sons Ltd, 2012.
- [104] W. Smith, J. Hashemi, *Foundations of Materials Science and Engineering*, fourth ed., McGraw-Hill Higher Education, 2006.
- [105] W.D. Callister, *Fundamentals of materials science and engineering: an interactive e-text*, fifth ed.s., John Wiley & Sons, Inc, 2001.
- [106] A.K. Jena, M.C. Chaturvedi, The role of alloying elements in the design of nickel-base superalloys, *J. Mater. Sci.* 19 (1984) 3121–3139.
- [107] C. Cui, Y. Gu, H. Harada, A. Sato, Microstructure and yield strength of UDIMET 720LI alloyed with Co–16.9 Wt pct Ti, *Metall. Mater. Trans.* 36 (2005) 2921–2927.
- [108] M.V. Nathal, R.D. Maier, L.J. Ebert, The influence of cobalt on the microstructure of the nickel-base superalloy MAR-M247, *Metall. Mater. Trans.* 13 (1982) 1775–1783.
- [109] L. Delehouzee, A. Deruyttere, The stacking fault density in solid solutions based on copper, silver, nickel, aluminium and lead, *Acta Mater.* 15 (1967) 727–734.
- [110] P.C.J. Gallagher, The influence of alloying, temperature, and related effects on the stacking fault energy, *Metall. Mater. Trans. B* 1 (1970) 2429–2461.
- [111] E.H. Köster, A.R. Thölen, A. Howie, Stacking fault energies of Ni–Co–Cr alloys, *Philos. Mag. A* (1964) 1093–1095.
- [112] B.E.P. Beeston, I.L. Dillamore, R.E. Smallman, The stacking-fault energy of some nickel–cobalt alloys, *Met. Sci.* 2 (1968) 12–14.
- [113] Y. Yuan, Y. Gu, C. Cui, T. Osada, Influence of Co content on stacking fault energy in Ni–Co base disk superalloys, *J. Mater. Res.* 26 (2011) 2833–2837.
- [114] J.Y. Chen, Q. Feng, L.M. Cao, Z.Q. Sun, Improvement of stress-rupture property by Cr addition in Ni-based single crystal superalloys, *Mater. Sci. Eng.* 528 (2011) 3791–3798.
- [115] X.X. Yu, C.Y. Wang, The effect of alloying elements on the dislocation climbing velocity in Ni: a first-principles study, *Acta Mater.* 57 (2009) 5914–5920.
- [116] N. Tsuno, K. Kakehi, C.M.F. Rae, R. Hashizume, Effect of ruthenium on creep strength of ni-base single-crystal superalloys at 750° C and 750 MPa, *Metall. Mater. Trans.* 40 (2008) 269–272.
- [117] Y. Ustinovshikov, Phase transition 'ordering-phase separation'in alloys of Ni–Cr system, *Mater. Chem. Phys.* 165 (2015) 14–18.

- [118] A. Kalogeridis, M. Kolbe, E. Nembach, The effect of chromium on the solubility limits of the γ' -forming elements in a Ni-Fe-Cr base superalloy, *Scr. Mater.* 31 (1994) 1239–1242.
- [119] Y. Luo, Y. Zhou, J. Liu, Influence of Ru and Cr on solidification process in a Re-free Ni-based single crystal superalloy, *Acta Metall. Sin.* 50 (2014) 1025–1030.
- [120] X. Gizan, E. Liu, Z. Zheng, Y. Yu, J. Tong, Y. Zhai, Solidification behavior and segregation of Re-containing cast Ni-base superalloy with different Cr content, *J. Mater. Sci. Technol.* 27 (2011) 113–117.
- [121] R.M. Kearsey, J.C. Beddoes, K.M. Jaansalu, W.T. Thompson, P. Au, The effects of Re, W and Ru on microsegregation behaviour in single crystal superalloy systems, *Superalloys 2004* (2004) 801–810.
- [122] W. Loomis, J. Freeman, D. Sponseller, The influence of molybdenum on the γ/γ' phase in experimental nickel-base superalloys, *Metall. Mater. Trans. B* 3 (1972) 989–1000.
- [123] P. Strunz, D. Mukherji, R. Gilles, Determination of γ' solution temperature in Re-rich Ni-base superalloy by small-angle neutron scattering, *J. Appl. Cryst.* 34 (2001) 541–548.
- [124] W. Qiong, L. Shu-Suo, M. Yue, G. Sheng-Kai, First principles calculations of alloying element diffusion coefficients in Ni using the five-frequency model, *Chin. Phys. B* 21 (2012) 1–7, 109102.
- [125] M. Krčmar, C.L. Fu, A. Janotti, R.C. Reed, Diffusion rates of 3d transition metal solutes in nickel by first-principles calculations, *Acta Mater.* 53 (2005) 2369–2376.
- [126] A. Janotti, M. Krčmar, C.L. Fu, R.C. Reed, Solute diffusion in metals: larger atoms can move faster, *Phys. Rev. Lett.* 92 (2004) 085901.
- [127] Z. Shi, J. Li, S. Liu, Effects of Ru on the microstructure and phase stability of a single crystal superalloy, *Int. J. Mach. Mach. Mater.* 19 (2012) 1004–1009.
- [128] F. Sun, J. Zhang, S. Mao, Y. Jiang, Q. Feng, Z. Shen, Kink structures induced in nickel-based single crystal superalloys by high-Z element migration, *J. Alloy. Comp.* 618 (2015) 750–754.
- [129] W.Z. Wang, T. Jin, J.L. Liu, X.F. Sun, H.R. Guan, Z.Q. Hu, Role of Re and Co on microstructures and γ' coarsening in single crystal superalloys, *Mater. Sci. Eng.* 479 (2008) 148–156.
- [130] J. Yu, X. Sun, N. Zhao, T. Jin, H. Guan, Z. Hu, Effect of heat treatment on microstructure and stress rupture life of DD32 single crystal Ni-base superalloy, *Mater. Sci. Eng.* 460–461 (2007) 420–427.
- [131] M. Huang, Z. Cheng, J. Xiong, J. Li, J. Hu, Z. Liu, J. Zhu, Coupling between Re segregation and γ/γ' interfacial dislocations during high-temperature, low-stress creep of a nickel-based single-crystal superalloy, *Acta Mater.* 76 (2014) 294–305.
- [132] P. Henderson, L. Berglin, C. Jansson, On rafting in a single crystal nickel-base superalloy after high and low temperature creep, *Scr. Mater.* 40 (1998) 229–234.
- [133] Z. Bi, J. Dong, L. Zheng, X. Xie, Phenomenon and mechanism of high temperature low plasticity in High-Cr nickel-based superalloy, *J. Mater. Sci. Technol.* 29 (2013) 187–192.
- [134] L. Zheng, C. Xiao, G. Zhang, G. Gu, D. Tang, Key factors in quality control of a high Cr content cast Ni-base superalloy K4648, *Rare Met.* 30 (2011) 410–413.
- [135] L. Zheng, C. Xiao, G. Zhang, B. Han, D. Tang, Primary α phase and its effect on the impact ductility of a high Cr content cast Ni-base superalloy, *J. Alloy. Comp.* 527 (2012) 176–183.
- [136] R.A. MacKay, T.P. Gabb, A. Garg, R.B. Rogers, Influence of composition on microstructural parameters of single crystal nickel-base superalloys, *Mater. Charact.* 70 (2012) 83–100.
- [137] K. Cheng, C. Jo, T. Jin, Z. Hu, Precipitation behavior of μ phase and creep rupture in single crystal superalloy CMSX-4, *J. Alloy. Comp.* 509 (2011) 7078–7086.
- [138] X.G. Liu, L. Wang, L.H. Lou, J. Zhang, Effect of Mo addition on microstructural characteristics in a Re-containing single crystal superalloy, *J. Mater. Sci. Technol.* 31 (2015) 143–147.
- [139] Y. Ru, S. Li, Y. Pei, J. Zhou, S. Gong, H. Xu, Interdendritic Mo homogenization and sub-solidus melting during solution treatment in the Mo-strengthening single crystal superalloys, *J. Alloy. Comp.* 662 (2016) 431–435.
- [140] V. Biss, G.N. Kirby, D.L. Sponseller, The relative effects of chromium, molybdenum, tungsten, and tantalum on the occurrence of σ phase in cast Ni-Co-Cr alloys, *Metall. Mater. Trans.* 7 (1976) 1251–1261.
- [141] S. Yang, M. Jiang, H. Li, Y. Liu, L. Wang, Assessment of Co-Cr-Ni ternary system by CALPHAD technique, *Rare Met.* 31 (2012) 75–80.
- [142] C.M.F. Rae, M.S.A. Karunaratne, C.J. Small, R.W. Broomfield, C.N. Jones, R.C. Reed, Topologically Close Packed Phases in an Experimental Rhenium-containing Single Crystal Superalloy, *Superalloy vol. 2000* (2000) 767–776.
- [143] K. Durst, A. Goken, Micromechanical characterisation of the influence of rhenium on the mechanical properties in nickel-base superalloys, *Mater. Sci. Eng.* 387 (2004) 312–316.
- [144] C.M.F. Rae, N. Matan, R.C. Reed, The role of stacking fault shear in the primary creep of [001]-oriented single crystal superalloys at 750°C and 750 MPa, *Mater. Sci. Eng.* 300 (2001) 125–134.
- [145] C.M.F. Rae, N. Matan, D.C. Cox, M.A. Rist, R.C. Reed, On the primary creep of CMSX-4 superalloy single crystals, *Metall. Mater. Trans.* 31 (2000) 2219–2228.
- [146] B. Seiser, R. Drautz, D.G. Pettifor, TCP phase predictions in Ni-based superalloys: structure maps revisited, *Acta Mater.* 59 (2011) 749–763.
- [147] K.A. Al-Jarba, G.E. Fuchs, Carbon-containing single-crystal nickel-based superalloys: segregation behavior and carbide formation, *JOM* 56 (2004) 50–55.
- [148] L.R. Liu, T. Jin, N.R. Zhao, Z.H. Wang, X.F. Sun, Effect of carbon addition on the creep properties in a Ni-based single crystal superalloy, *Mater. Sci. Eng.* 385 (2004) 105–112.
- [149] Y. Zhou, A. Volek, Effect of carbon additions on hot tearing of a second generation nickel-base superalloy, *Mater. Sci. Eng.* 479 (2008) 324–332.
- [150] Z. Shi, J. Li, S. Liu, Effect of Hf on stress rupture properties of DD6 single crystal superalloy after long term aging, *J. Iron Steel Res. Int.* 19 (2012) 66–70.
- [151] Y. Xu, Q. Jin, X. Xiao, X. Cao, K.S. Song, Strengthening mechanisms of carbon in modified nickel-based superalloy Nimonic 80A, *Mater. Sci. Eng.* 528 (2011) 4600–4607.
- [152] S. Xiang, S. Mao, Z. Shen, H. Long, H. Wei, S. Ma, J. Zhang, Y. Chen, J. Zhang, B. Zhang, Y. Liu, Site preference of metallic elements in M₂₃C₆ carbide in a Ni-based single crystal superalloy, *Mater. Des.* 129 (2017) 9–14.
- [153] L. Wang, D. Wang, T. Liu, X.W. Li, W.G. Jiang, Effect of minor carbon additions on the high-temperature creep behavior of a single-crystal nickel-based superalloy, *Mater. Charact.* 104 (2015) 81–85.
- [154] H.S. Lee, D.S. Kim, K.B. Yoo, K.S. Song, Quantitative analysis of carbides and the sigma phase in thermally exposed GTD-111, *Met. Mater. Int.* 18 (2012) 287–293.
- [155] J.X. Yang, Q. Zheng, X.F. Sun, H.R. Guan, Z.Q. Hu, Formation of μ phase during thermal exposure and its effect on the properties of K465 superalloy, *Scr. Mater.* 55 (2006) 331–334.
- [156] L.R. Liu, T. Jin, N.R. Zhao, Z.H. Wang, X.F. Sun, H.R. Guan, Effect of carbon additions on the microstructure in a Ni-base single crystal superalloy, *Mater. Lett.* 58 (2004) 2290–2294.
- [157] Z. Yu, X. Zhao, W. Zhang, J. Zhang, F. Zhi, Effect of carbon additions on the microstructure of a single crystal Ni-based superalloy AM3, *China Foundry* 7 (2010) 352–356.
- [158] A. Epishin, U. Brückner, P.D. Portella, T. Link, Influence of small rhenium additions on the lattice spacing of nickel solid solution, *Scr. Mater.* 48 (2003) 455–459.
- [159] A. Heckl, S. Neumeier, S. Cenancovic, M. Göken, R.F. Singer, Reasons for the enhanced phase stability of Ru-containing nickel-based superalloys, *Acta Mater.* 59 (2011) 6563–6573.
- [160] C. Schulze, M. Feller-Kniepmeier, Transmission electron microscopy of phase composition and lattice misfit in the Re-containing nickel-base superalloy CMSX-10, *Mater. Sci. Eng.* 281 (2000) 204–212.
- [161] S.G. Tian, M.G. Wang, H.C. Yu, X.F. Yu, T. Li, B.J. Qian, Influence of element Re on lattice misfits and stress rupture properties of single crystal nickel-based superalloys, *Mater. Sci. Eng.* 527 (2010) 4458–4465.
- [162] R.A. Mackay, M.V. Nathal, D.D. Pearson, Influence of molybdenum on the creep properties of nickel-base superalloy single crystals, *Metall. Mater. Trans.* 21 (1990) 381–388.
- [163] T.C. Tiearney, N.J. Grant, Measurement of structural parameters important in creep of Ni-Mo and Ni-W solid solutions, *Metall. Mater. Trans.* 13 (1982) 1827–1836.
- [164] J. Chen, Q. Feng, L. Cao, Z. Sun, Effects of Ru and Cr on γ/γ' microstructural evolution of Ni-Based single crystal superalloys during heat treatment, *Acta Metall. Sin.* 46 (2010) 897–906.
- [165] W.F. Hosford, *Mechanical Behavior of Materials*, Cambridge University, 2005.
- [166] A.J. Ardell, R.B. Nicholson, On the modulated structure of aged Ni-Al alloys: with an Appendix on the elastic interaction between inclusions by J. D. Eshelby, *Acta Mater.* 10 (1966) 1295–1309.
- [167] J.D. Eshelby, The determination of the elastic field of an ellipsoidal inclusion, and related problems, *Proc. Roy. Soc. Lond.* 241 (1957) 376–396.
- [168] A. Baldan, Adhesively-bonded joints and repairs in metallic alloys, polymers and composite materials: adhesives, adhesion theories and surface pre-treatment, *J. Mater. Sci.* 39 (2002) 1–49.
- [169] N. Matan, D.C. Cox, C.M.F. Rae, R.C. Reed, Creep of CMSX-4 superalloy single crystals: effects of rafting at high temperature, *Acta Mater.* 47 (1999) 3367–3381.
- [170] F.R.N. Nabarro, The chemical driving force for rafting in superalloys, *Scr. Mater.* 37 (1997) 497–501.
- [171] F.R.N. Nabarro, C.M. Cress, P. Kotschy, The thermodynamic driving force for rafting in superalloys, *Acta Mater.* 44 (1996) 3189–3198.
- [172] J.S. Van Sluytman, T.M. Pollock, Optimal precipitate shapes in nickel-base $\gamma-\gamma'$ alloys, *Acta Mater.* 60 (2012) 1771–1783.
- [173] U.G. Rainer Völkl, M. Feller-Kniepmeier, Measurement of the unconstrained misfit in the nickel-base superalloy CMSX-4 with CBED, *Scr. Mater.* 38 (1998) 893–900.
- [174] A. Royer, P. Bastie, M. Veron, In situ determination of γ' phase volume fraction and of relations between lattice parameters and precipitate morphology in Ni-based single crystal superalloy, *Acta Mater.* 46 (1998) 5357–5368.
- [175] C. Schulze, M. Feller-Kniepmeier, Transmission electron microscopy of phase composition and lattice misfit in the Re-containing nickel-base superalloy CMSX-10, *Mater. Sci. Eng.* 281 (2000) 204–212.
- [176] T.M. Pollock, A.S. Argon, Creep resistance of CMSX-3 nickel base superalloy single crystals, *Acta Mater.* 40 (1992) 1–30.
- [177] J.X. Zhang, T. Murakumo, H. Harada, Y. Koizumi, T. Kobayashi, Creep deformation mechanisms in some modern single-crystal superalloys, *Superalloys 2004* (2004) 189–195.
- [178] J.X. Zhang, T. Murakumo, H. Harada, Y. Koizumi, Dependence of creep strength on the interfacial dislocations in a fourth generation SC superalloy

- TMS-138, *Scr. Mater.* 48 (2003) 287–293.
- [179] H. Long, H. Wei, Y. Liu, S. Mao, J. Zhang, S. Xiang, Y. Chen, W. Gui, Q. Li, Z. Zhang, X. Han, Effect of lattice misfit on the evolution of the dislocation structure in Ni-based single crystal superalloys during thermal exposure, *Acta Mater.* 120 (2016) 95–107.
- [180] H. Long, S. Mao, S. Xiang, Y. Chen, H. Wei, Y. Zhou, J. Liu, Y. Liu, A modification on Brook formula in calculating the misfit of Ni-based superalloys, *Mater. Des.* 126 (2017) 12–17.
- [181] L.J. Carroll, Q. Feng, T.M. Pollock, Interfacial dislocation networks and creep in directional coarsened Ru-containing nickel-base single-crystal superalloys, *Metall. Mater. Trans.* 39 (2008) 1290–1307.
- [182] A. Lasalmonie, J.L. Strudel, Interfacial dislocation networks around γ' precipitates in nickel-base alloys, *Phil. Mag.* 32 (1975) 937–949.
- [183] Y. Koizumi, T. Kobayashi, T. Yokokawa, J.X. Zhang, M. Osawa, H. Harada, Y. Aoki, M. Arai, Development of next-generation Ni-base single crystal superalloys, *Superalloys 2004* (2004) 35–43.
- [184] J.X. Zhang, Y. Koizumi, T. Kobayashi, T. Murakumo, H. Harada, *Metall. Mater. Trans.* 35 (2004) 1911–1913.
- [185] L.J. Carroll, Q. Feng, J.F. Mansfield, T.M. Pollock, Elemental partitioning in Ru-containing nickel-base single crystal superalloys, *Mater. Sci. Eng.* 457 (2007) 292–299.
- [186] A. Heckl, S. Neumeier, S. Cenancovic, M. Göken, Reasons for the enhanced phase stability of Ru-containing nickel-based superalloys, *Acta Mater.* 59 (2011) 6563–6573.
- [187] X. Lv, J. Zhang, Q. Feng, The promotion of Ru on topologically close-packed phase precipitation in the high Cr-containing (~9wt.%) nickel-base single crystal superalloy, *J. Alloy. Comp.* 648 (2015) 853–857.
- [188] Z. Shi, J. Li, S. Liu, Effects of Ru on the microstructure and phase stability of a single crystal superalloy, *Int. J. Mach. Mach. Mater.* 19 (2012) 1004–1009.
- [189] Z. Shi, J. Li, S. Liu, X. Wang, X. Yue, Effect of Ru on stress rupture properties of nickel-based single crystal superalloy at high temperature, *Trans. Nonferrous Metals Soc. China* 22 (2012) 2106–2111.
- [190] X.G. Wang, J.L. Liu, T. Jin, X.F. Sun, The effects of ruthenium additions on tensile deformation mechanisms of single crystal superalloys at different temperatures, *Mater. Des.* 63 (2014) 286–293.
- [191] K. Matuszewski, R. Rettig, H. Matysiak, Z. Peng, I. Povstugar, P. Choi, J. Müller, D. Raabe, E. Spiecker, K.J. Kurzydowski, R.F. Singer, Effect of ruthenium on the precipitation of topologically close packed phases in Ni-based superalloys of 3rd and 4th generation, *Acta Mater.* 95 (2015) 274–283.
- [192] R. Rettig, A. Heckl, R.F. Singer, Modeling of precipitation kinetics of TCP-phases in single crystal nickel-base superalloys, *Adv. Mater. Res.* 278 (2011) 180–185.
- [193] R. Rettig, R.F. Singer, Influence of Ruthenium on topologically close packed phase precipitation in single-crystal Ni-based superalloys: numerical experiments and validation, *Superalloys 2012* (2012) 205–214.
- [194] R.J. Bennett, R. Krakow, A.S. Eggeman, C.N. Jones, H. Murakami, C.M.F. Rae, On the oxidation behavior of titanium within coated nickel-based superalloys, *Acta Mater.* 92 (2015) 278–289.
- [195] M. Bensch, J. Preußner, R. Hüttner, G. Obigodi, S. Virtanen, J. Gabel, U. Glatzel, Modelling and analysis of the oxidation influence on creep behaviour of thin-walled structures of the single-crystal nickel-base superalloy René N5 at 980 °C, *Acta Mater.* 58 (2010) 1607–1617.
- [196] N.P. Padture, K.W. Schlichting, T. Bhatia, A. Ozturk, B. Cetegen, E.H. Jordan, M. Gell, S. Jiang, T.D. Xiao, P.R. Strutt, E. Garcia, P. Miranzo, M.I. Osendi, Towards durable thermal barrier coatings with novel microstructures deposited by solution-precursor plasma spray, *Acta Mater.* 49 (2001) 2251–2257.
- [197] Y.C. Zhou, T. Hashida, Coupled effects of temperature gradient and oxidation on thermal stress in thermal barrier coating system, *Int. J. Solid Struct.* 38 (2001) 4235–4264.
- [198] E.P. Busso, J. Lin, S. Sakurai, M. Nakayama, A mechanistic study of oxidation-induced degradation in a plasma-sprayed thermal barrier coating system: Part I: model formulation, *Acta Mater.* 49 (2001) 1515–1528.
- [199] M.H. Li, Z.Y. Zhang, X.F. Sun, H.R. Guan, W.Y. Hu, Z.Q. Hu, Oxidation and degradation of EB-PVD thermal-barrier coatings, *Oxid. Metals* 58 (2002) 499–512.
- [200] U. Schulz, M. Menzebach, C. Leyens, Y.Q. Yang, Influence of substrate material on oxidation behavior and cyclic lifetime of EB-PVD TBC systems, *Surf. Coating. Technol.* 146–147 (2001) 117–123.
- [201] H.R. Guan, M.H. Li, X.F. Sun, Z.Y. Zhang, W.Y. Hu, S.K. Gong, T. Jin, N.R. Zhao, Z.Q. Hu, Investigation on oxidation and failure of the thermal barrier coating deposited on superalloy, *Acta Metall. Sin.* 38 (2002) 1133–1140.
- [202] Y. Song, H. Murakami, C. Zhou, Cyclic-oxidation behavior of multilayered Pt/Ru-modified aluminide coating, *J. Mater. Sci. Technol.* 27 (2011) 280–288.
- [203] S. Widjaja, A.M. Limarga, T.H. Yip, Oxidation behavior of a plasma-sprayed functionally graded ZrO_2/Al_2O_3 thermal barrier coating, *Mater. Lett.* 57 (2002) 628–634.
- [204] C.T. Liu, J. Ma, X.F. Sun, Oxidation behavior of a single-crystal Ni-base superalloy between 900 and 1000 °C in air, *J. Alloy. Comp.* 491 (2010) 522–526.
- [205] Y. Wu, T. Narita, Oxidation behavior of the single crystal Ni-based superalloy at 900 °C in air and water vapor, *Surf. Coating. Technol.* 202 (2007) 140–145.
- [206] K. Kawagishi, H. Harada, A. Sato, A. Sato, T. Kobayashi, The oxidation properties of fourth generation single-crystal nickel-based superalloys, *JOM* 58 (2006) 43–46.
- [207] M.H. Li, X.F. Sun, J.G. Li, Z.Y. Zhang, T. Jin, H.R. Guan, Oxidation behavior of a single-crystal Ni-base superalloy in air. I: at 800 and 900 °C, *Oxid. Met.* 59 (2003) 591–605.
- [208] N. Birks, G.H. Meier, F.S. Pettit, Forming continuous alumina scales to protect superalloys, *JOM* 46 (1994) 42–46.
- [209] A.D. Celal, D.N. Duhl, Second-generation nickel-base single crystal superalloy, *Superalloys 1988* (1988) 235–244.